

MAGNITUDE OF MODULE CATEGORIES AND SPECIAL BISERIAL ALGEBRAS

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ABSTRACT. Børve, Horiatakis, and Kalck defined the magnitude of the module category of a representation-finite algebra as the sum of the entries of the inverse of the matrix of Hom dimensions between indecomposable modules. They showed it is an Euler characteristic, the alternating count of vertices, arrows, and meshes of the Auslander–Reiten quiver, and conjectured that it is at least the number of simple modules, with equality precisely for special biserial algebras. We prove this conjecture over an algebraically closed field. For a representation-directed algebra, we relate the magnitudes of the module category, the subcategory of modules without a given simple composition factor, and the ideal quotient by it. By the theory of hammocks of Ringel and Vossieck, this quotient is equivalent to a category of representations of a finite poset, whose magnitude we compute. The general case reduces to the representation-directed case through algebras obtained by grading a standard form.

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1. INTRODUCTION

Magnitude is an invariant of enriched categories introduced by Leinster [19, Section 1.3]; for finite categories it generalizes the Euler characteristic of the classifying space [18]. Chuang, King, and Leinster studied the magnitude of finite-dimensional algebras [8]. Børve, Horiatakis, and Kalck recently defined the magnitude of the module category of a representation-finite algebra A as the magnitude of its Auslander algebra [6]. Over an algebraically closed field k , it is

$$\chi(\text{mod } A) = \mathbf{1}^\top H_A^{-1} \mathbf{1}, \quad H_A = (\dim_k \text{Hom}_A(X, Y))_{X, Y \in \text{ind } A},$$

the sum of the entries of the inverse of the invertible Hom matrix between the indecomposable modules. They showed that it is the Euler characteristic of the Auslander–Reiten quiver Γ_A : the alternating sum of the numbers of vertices, arrows, and meshes, which correspond to cells of dimensions 0, 1, and 2 [6, Lemma 1.9]:

$$\chi(\text{mod } A) = |\text{ind } A| - a(A) + (|\text{ind } A| - |A|). \quad (1.1)$$

Here $a(A)$ is the number of arrows of Γ_A , and $|A|$ is the number of simple A -modules up to isomorphism, so that $|\text{ind } A| - |A|$ counts the meshes, one for each nonprojective indecomposable.

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Børve, Horiatakis, and Kalck computed this invariant for several classes of algebras. It equals $|A|$ for representation-finite special biserial algebras [6, Theorem 3.5], and $h_Q - 1$ for the path algebra of a Dynkin quiver Q with Coxeter number h_Q [6, Theorem 4.3]. They conjectured that $\chi(\text{mod } A) \geq |A|$ for every representation-finite bound path algebra over a field, with equality precisely for special biserial algebras [6, Conjecture A]. They proved this for representation-finite hereditary algebras (a consequence of [6, Theorem 4.3]), radical square zero algebras [6, Corollary 4.6], and self-injective algebras [6, Theorem 5.8 and Corollary 5.9]. We prove the conjecture over an algebraically closed field.

Theorem A (= Theorem 6.8). *Let A be a finite-dimensional representation-finite algebra over an algebraically closed field. Then the following hold.*

- (1) $\chi(\text{mod } A) \geq |A|$.
- (2) $\chi(\text{mod } A) = |A|$ if and only if A is special biserial.

By (1.1), Theorem A is a statement about the shape of Auslander–Reiten quivers: the number of arrows is at most twice the number of nonprojective indecomposables, with equality precisely for special biserial algebras. We call the difference

$$\sigma(A) = \chi(\text{mod } A) - |A| = 2(|\text{ind } A| - |A|) - a(A)$$

the *magnitude surplus*.

Strategy of the proof. The proof first treats representation-directed algebras by removing one simple module at a time. For a primitive idempotent e of A , we call the algebra A/AeA the *primitive deletion* of e . For a basic, representation-finite, and representation-directed algebra A , we show that

$$\sigma(A/AeA) \leq \sigma(A),$$

with equality only if $\dim_k Xe \leq 1$ for every indecomposable A -module X (Proposition 4.19).

This comes from a general identity (Theorem 3.12). For an idempotent e of A , put $B = A/AeA$, and let $\mathcal{D} = (\text{mod } A)/[\text{mod } B]$ be the *ideal quotient* by the maps factoring through $\text{mod } B$. Like $\chi(\text{mod } A)$, the magnitude of $\mathcal{D}$ is the Euler characteristic of its translation quiver $\Gamma_{\mathcal{D}}$ (Proposition 2.14 and Definition 4.10), and

$$\chi(\text{mod } A) - \chi(\text{mod } B) - \chi(\mathcal{D}) = c - r - \delta,$$

where c is the sum, over pairs of indecomposable B -modules, of the difference between the numbers of arrows between them in Γ_B and in Γ_A , r is the number of almost split sequences of $\text{mod } B$ that are not almost split in $\text{mod } A$, and $\delta \geq 0$ counts, for each almost split sequence of $\text{mod } A$ with exactly one end term in $\text{mod } B$, all but one of its middle summands outside $\text{mod } B$. When A is representation-finite and representation-directed and e is primitive, $\delta = 0$ and $r \leq c$; since $|A| = |B| + 1$, it then suffices to show $\chi(\mathcal{D}) \geq 1$. In this case Ringel and Vossieck identified $\mathcal{D}$ with the category $T\text{-Sp}$ of representations of a finite poset T by subspaces [21, Section 10, Theorem, p. 243]: such a representation is a vector space W , its *total space*, with subspaces W_t for $t \in T$ such that $W_s \subseteq W_t$ whenever $s \leq t$. We compute the magnitude of these categories.

Theorem B (= Theorem 4.5). *Let T be a finite poset such that $T\text{-Sp}$ is representation-finite. In the Auslander–Reiten quiver of $T\text{-Sp}$, all paths from the representation k with all subspaces zero to the representation k with all subspaces k have the same length L , and*

$$\chi(T\text{-Sp}) = 1 + L - |T| \geq 1.$$

Equality holds if and only if T has width at most two, if and only if every indecomposable has one-dimensional total space.

The inequality $L \geq |T|$ and the equivalence of $L = |T|$ with the two conditions of Theorem B are due to Ringel and Vossieck [21, Section 8, p. 237]; the new part of Theorem B is the formula for $\chi(T\text{-Sp})$. For a basic, representation-finite, and representation-directed algebra A , deleting simple modules one at a time then gives $\sigma(A) \geq 0$. If $\sigma(A) = 0$, then $\sigma(A/AeA) = \sigma(A)$ for every primitive idempotent e , so $\dim_k Xe \leq 1$ for all e and all indecomposable X . Thus every indecomposable A -module is thin, that is, has each simple module at most once as a composition

factor. Hence A is biserial by Proposition 4.24, due to Pogorzały and Skowroński, and special biserial by Theorem 4.25, due to Skowroński and Waschbüsch.

For a general A , we pass to a standard form S of A [5, 7]: an algebra with the same Auslander–Reiten quiver, whose indecomposable modules and maps between them are given by paths in this quiver modulo the mesh relations. Path length grades the Hom spaces between indecomposable S -modules, and hence S . The graded S -modules in degrees $0, \dots, m$ are the modules over a representation-directed algebra S_m , which we call an interval algebra. One has $\sigma(S_m) = (m+1)\sigma(A) + O(1)$ as $m \rightarrow \infty$ (Theorem 5.5, with $\sigma(S) = \sigma(A)$), and $\sigma(S_m) \geq 0$ by Corollary 4.26; hence $\sigma(A) \geq 0$ (Section 5).

Now suppose $\sigma(A) = 0$. The interval algebra S_m is block upper triangular, with one diagonal block for each degree in $[0, M]$. For every $q \geq 1$ and suitable M , deleting the degrees in the gaps between q well separated copies of $[0, m]$, that is, factoring out their diagonal idempotents, leaves a product of q copies of S_m . The surplus is additive on products and does not increase under deletion, so $q\sigma(S_m) \leq \sigma(S_m)$, and $\sigma(S_m) = O(1)$ as q grows because $\sigma(A) = 0$. Hence $\sigma(S_m) = 0$ for every m (Lemma 6.1), and every S_m is biserial by Corollary 4.26. By results of Auslander and Reiten and of Skowroński and Waschbüsch, a representation-finite algebra is biserial if and only if the middle term of every almost split sequence has at most two nonprojective indecomposable summands (Theorem 6.4). The almost split sequences of $\text{mod } S$ lift to almost split sequences over S_m for large m , with the same nonprojective middle summands, and A and S have isomorphic Auslander–Reiten quivers. Hence A is biserial, and special biserial by Theorem 4.25 (Section 6). Figure 1 summarizes these steps.

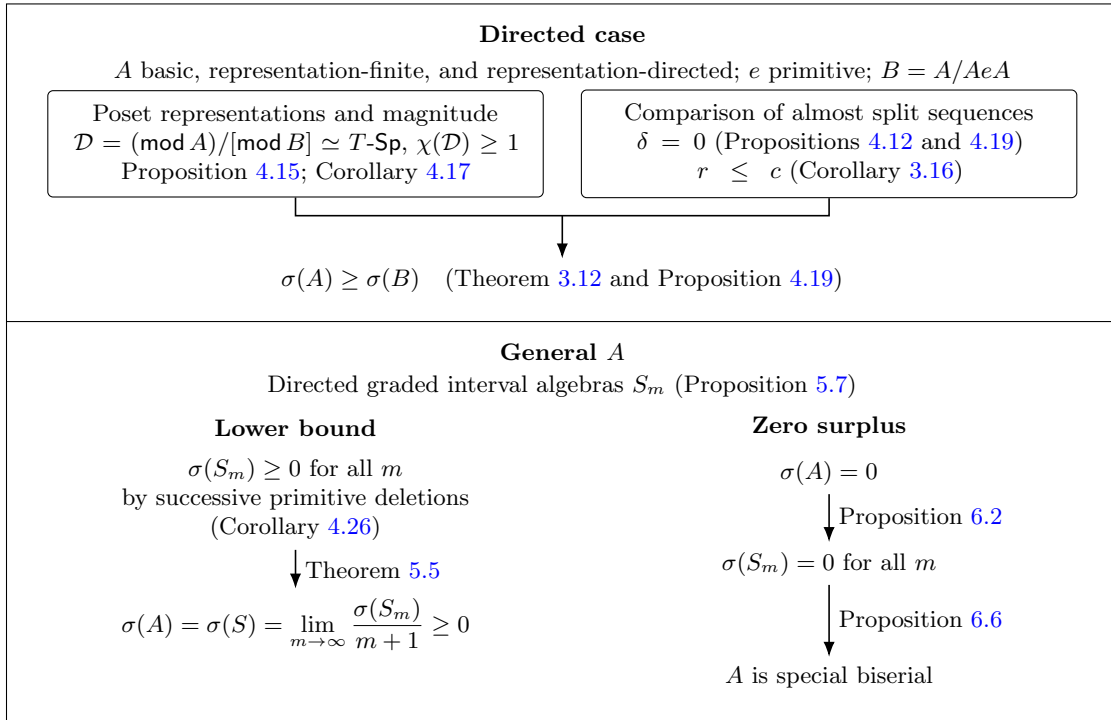


FIGURE 1. Primitive deletion and the two parts of Theorem A. Here $\sigma(A) = \chi(\text{mod } A) - |A|$, $T\text{-Sp}$ denotes representations of a finite poset T by subspaces, S is a standard form of A with $\sigma(S) = \sigma(A)$ (Theorem 5.8 and Lemma 5.10), and $\text{mod } S_m$ is equivalent to the category of graded S -modules supported in degrees $0, \dots, m$ (Proposition 5.4).

Organization. Section 2 recalls almost split sequences and Iyama’s τ -categories, and extends the proof of (1.1) in [6] to strict τ -categories. Section 3 proves the identity of Theorem 3.12 for the ideal

quotient by a Serre subcategory and a sufficient condition for the inequality $r \leq c$. Section 4 proves Theorem B and the inequality $\sigma(A/AeA) \leq \sigma(A)$ for a primitive idempotent e . Section 5 defines interval algebras and proves Theorem A(1), and Section 6 proves Theorem A(2). Appendix A gives direct proofs of the three results on the ideal quotient in Section 4 that use [21], so that Theorem A does not depend on [21]; Theorem B for an arbitrary poset does.

Conventions and notation. Unless specified otherwise, k is algebraically closed and algebras are finite-dimensional k -algebras. Modules are finite-dimensional right modules, and $\text{mod } A$ denotes their category. Composition is written gf for first f , then g . We use the same order for paths: $\beta\alpha$ is the path α followed by β . For a finite quiver Q , right modules over $A = (kQ)^{\text{op}}$, or over its quotient by a specified ideal, are identified with representations of Q : the space at a vertex i is Me_i , and an arrow $\alpha : i \rightarrow j$ acts as a map $Me_i \rightarrow Me_j$. We write P_i , I_i , and S_i for the indecomposable projective, injective, and simple modules at the vertex i . For a module M , $|M|$ is the number of isomorphism classes of indecomposable direct summands of M ; thus $|A| = |A_A|$ is the number of isomorphism classes of simple modules. We write $D_k = \text{Hom}_k(-, k)$, and add M for direct summands of finite direct sums of copies of M . For a Krull–Schmidt category $\mathcal{C}$, $\text{ind } \mathcal{C}$ is a chosen set of indecomposable representatives, also regarded as the full subcategory on those objects. We write $\mathcal{C}(X, Y) = \text{Hom}_{\mathcal{C}}(X, Y)$ and abbreviate $\text{ind}(\text{mod } A)$ to $\text{ind } A$. A category is representation-finite if it has finitely many indecomposable isomorphism classes. The categorical radical is $\text{rad}_{\mathcal{C}}$, and $\text{Irr}_{\mathcal{C}}(X, Y) = \text{rad}_{\mathcal{C}}(X, Y) / \text{rad}_{\mathcal{C}}^2(X, Y)$. The *irreducible quiver* of $\mathcal{C}$ has vertices $\text{ind } \mathcal{C}$ and $\dim_k \text{Irr}_{\mathcal{C}}(X, Y)$ arrows from X to Y . We write $\text{indeg}_{\mathcal{C}}(X)$ and $\text{outdeg}_{\mathcal{C}}(X)$ for the numbers of incoming and outgoing arrows, including multiplicities. The *Auslander–Reiten quiver* Γ_A is this quiver of $\text{mod } A$, equipped with Auslander–Reiten translation and with its projective and injective vertices marked. Translations are displayed by dotted arrows $X \dashrightarrow \tau_A X$; they do not count as arrows in paths or arrow counts. An algebra is *representation-directed* if there is no cycle of nonzero nonisomorphisms between indecomposable modules. We also use *directed* in this sense for an algebra and for a category. Special biserial and biserial algebras are recalled in Definitions 4.22 and 4.23; a non-basic algebra is special biserial if its basic algebra is.

Formalization. A Lean 4 formalization of Theorem A is available at <https://github.com/haruhisa-enomoto/magnitude-conjecture>. It follows the proof in this paper, with the arguments of Appendix A.

Use of AI. The research was done under the author’s direction using GPT-5.6-Sol, Claude Fable 5, Claude Opus 5, and GPT-6-Astra. GPT-5.6-Sol and Claude Fable 5 developed the initial proof, and GPT-5.6-Sol produced earlier manuscript drafts. GPT-6-Astra subsequently developed the simplified proof presented here and wrote a new manuscript based on it, keeping selected arguments and examples from the earlier drafts. It also produced the Lean formalization. Claude Opus 5 extensively reviewed the proofs and successive manuscript drafts. The author directed the research and revisions and set the structure and much of the wording of the present text. The author is responsible for the mathematical content and the final manuscript.

2. PRELIMINARIES ON τ -CATEGORIES AND MAGNITUDE

We recall almost split sequences and their generalization to additive categories. Each such sequence gives the beginning of a projective resolution of a simple functor. Following the proof of (1.1) in [6, Lemma 1.9], these resolutions give an analogue of (1.1) (Proposition 2.14) for strict τ -categories (Definition 2.12), in which the resolutions have length two. Module categories and the ideal quotients used below are strict τ -categories.

2.1. Almost split maps and τ -sequences. We recall almost split maps and sequences, then define the τ -sequences that replace them in ideal quotients.

Let $\mathcal{C}$ be a Krull–Schmidt k -linear category, for example $\text{mod } A$. A map $f : V \rightarrow X$ in $\mathcal{C}$ to an indecomposable X is *right almost split* if it is not a retraction and every non-retraction to X factors through it. It is *right minimal* if every $u \in \text{End}(V)$ with $fu = f$ is invertible. Left almost split maps and left minimality are defined dually, using sections.

Definition 2.1. Let A be an algebra. For a nonprojective indecomposable A -module X , write its almost split sequence as

$$0 \longrightarrow \tau_A X \xrightarrow{\nu_X^+} \theta_A X \xrightarrow{\mu_X^+} X \longrightarrow 0.$$

At a projective X , put $\tau_A X = 0$ and $\theta_A X = \text{rad } X$, and let μ_X^+ be the inclusion. Dually, write the sequence starting at a noninjective X as

$$0 \longrightarrow X \xrightarrow{\mu_X^-} \theta_A^- X \xrightarrow{\nu_X^-} \tau_A^- X \longrightarrow 0.$$

At an injective X , put $\tau_A^- X = 0$ and $\theta_A^- X = X/\text{soc } X$, and let μ_X^- be the quotient map. Extend $\tau_A, \tau_A^-, \theta_A, \theta_A^-$ additively to arbitrary modules, as assignments on isomorphism classes.

The following standard facts relate almost split maps to the arrows of Γ_A . For an indecomposable Y , write $[V : Y]_\oplus$ for its multiplicity as a direct summand of V .

Proposition 2.2 ([3, V, Section 1; VII, Proposition 1.3]). *Let X, Y be indecomposable A -modules. Then μ_X^+ is minimal right almost split and μ_X^- is minimal left almost split, and*

$$[\theta_A X : Y]_\oplus = \dim_k \text{Irr}_A(Y, X), \quad [\theta_A^- X : Y]_\oplus = \dim_k \text{Irr}_A(X, Y). \quad (2.1)$$

The map τ_A is a bijection from nonprojective to noninjective indecomposables, with inverse τ_A^- .

We will compute Ext groups from Hom groups by the following form of Auslander–Reiten duality.

Proposition 2.3 ([1, IV, Theorem 2.13]). *For modules M, N , there are natural isomorphisms*

$$D_k \text{Ext}_A^1(M, N) \cong \underline{\text{Hom}}_A(\tau_A^- N, M) \cong \overline{\text{Hom}}_A(N, \tau_A M). \quad (2.2)$$

Here $\underline{\text{Hom}}$ is Hom modulo maps through projectives, and $\overline{\text{Hom}}$ is Hom modulo maps through injectives. The formulas include the zero cases when N is injective or M is projective.

The ideal quotients studied in Section 3 are in general not abelian. In them, almost split sequences are replaced by Iyama’s τ -sequences, which we define through functors on the category.

Definition 2.4. Let $\mathcal{C}$ be a Hom-finite Krull–Schmidt k -linear category. Write $\text{mod } \mathcal{C}$ for the abelian category of contravariant k -linear functors from $\mathcal{C}$ to finite-dimensional vector spaces. For an object X , put $\mathbf{P}_X = \mathcal{C}(-, X)$, a projective object of $\text{mod } \mathcal{C}$ by the Yoneda lemma, and for a map $f : L \rightarrow M$ write $\mathbf{P}_f = f \circ - : \mathbf{P}_L \rightarrow \mathbf{P}_M$. For an indecomposable X , put

$$\mathbf{S}_X = \mathbf{P}_X / \text{rad}_{\mathcal{C}}(-, X).$$

It is simple, with $\mathbf{S}_X(Y) = 0$ for indecomposables $Y \not\cong X$, and $\mathbf{S}_X(X) = \text{End}_{\mathcal{C}}(X) / \text{rad}_{\mathcal{C}}(X, X) = k$, since this is a finite-dimensional division algebra over the algebraically closed field k . Dually, for covariant functors put $\mathbf{P}^X = \mathcal{C}(X, -)$, $\mathbf{P}^f = - \circ f : \mathbf{P}^M \rightarrow \mathbf{P}^L$, and $\mathbf{S}^X = \mathbf{P}^X / \text{rad}_{\mathcal{C}}(X, -)$.

Definition 2.5. A map $f : L \rightarrow M$ in $\mathcal{C}$ is a *weak kernel* of $g : M \rightarrow N$ if $\mathbf{P}_L \xrightarrow{\mathbf{P}_f} \mathbf{P}_M \xrightarrow{\mathbf{P}_g} \mathbf{P}_N$ is exact, that is, $gf = 0$ and every map $h : W \rightarrow M$ with $gh = 0$ factors through f . It is minimal if $fu = f$ forces $u \in \text{End}(L)$ to be invertible. Weak cokernels and their minimality are defined dually, using $\mathbf{P}^N \rightarrow \mathbf{P}^M \rightarrow \mathbf{P}^L$.

Definition 2.6 ([16, Section 2.1]; in the formulation of [15, Section 1.3]). Let $\mathcal{C}$ be as in Definition 2.4. A complex $L \xrightarrow{f} M \xrightarrow{g} N$ with f, g radical is a *right τ -sequence* if f is a minimal weak kernel of g and the sequences

$$\mathbf{P}_M \xrightarrow{\mathbf{P}_g} \text{rad}_{\mathcal{C}}(-, N) \longrightarrow 0, \quad \mathbf{P}^M \xrightarrow{\mathbf{P}^f} \text{rad}_{\mathcal{C}}(L, -) \longrightarrow 0$$

are exact. A *left τ -sequence* satisfies the same conditions, with g a minimal weak cokernel of f instead. The category $\mathcal{C}$ is a τ -category if every indecomposable is the last term of a right τ -sequence and the first term of a left τ -sequence.

The terms L and M of a right τ -sequence may be zero, and dually for left τ -sequences. The two exactness conditions say that every radical map into N factors through g , and every radical map out of L factors through f . For a right τ -sequence ending at an indecomposable N , the first condition and the weak kernel property give an exact sequence

$$P_L \xrightarrow{P_f} P_M \xrightarrow{P_g} P_N \longrightarrow S_N \longrightarrow 0. \quad (2.3)$$

Since f and g are radical and f is a minimal weak kernel, the sequence (2.3) is the beginning of a minimal projective resolution of S_N . Dually, a left τ -sequence starting at an indecomposable L gives the beginning of a minimal projective resolution $P^N \rightarrow P^M \rightarrow P^L \rightarrow S^L \rightarrow 0$.

Let $\mathcal{C}$ be a τ -category. For an indecomposable X , the right τ -sequence ending at X and the left τ -sequence starting at X are unique up to isomorphism of complexes [16, Section 2.1]. Following [15, Section 1.3], we write them as

$$\tau_{\mathcal{C}}^+ X \xrightarrow{\nu_X^+} \theta_{\mathcal{C}}^+ X \xrightarrow{\mu_X^+} X, \quad X \xrightarrow{\mu_X^-} \theta_{\mathcal{C}}^- X \xrightarrow{\nu_X^-} \tau_{\mathcal{C}}^- X.$$

We call X τ -projective if $\tau_{\mathcal{C}}^+ X = 0$, and τ -injective if $\tau_{\mathcal{C}}^- X = 0$. Following [16, Section 2.1(3)], we denote the sets of τ -projective and τ -injective objects of $\text{ind } \mathcal{C}$ by $\text{ind}_1^+ \mathcal{C}$ and $\text{ind}_1^- \mathcal{C}$. We use the following properties, proved by Iyama.

Proposition 2.7 ([16, Sections 2.1 and 2.3]). *Let $\mathcal{C}$ be a τ -category, and let X be an indecomposable.*

- (1) *The map μ_X^+ is minimal right almost split, and $\tau_{\mathcal{C}}^+ X$ is zero or indecomposable. Dually, μ_X^- is minimal left almost split, and $\tau_{\mathcal{C}}^- X$ is zero or indecomposable.*
- (2) *If $\tau_{\mathcal{C}}^+ X \neq 0$, then $\tau_{\mathcal{C}}^+ X \xrightarrow{\nu_X^+} \theta_{\mathcal{C}}^+ X \xrightarrow{\mu_X^+} X$ is also the left τ -sequence starting at $\tau_{\mathcal{C}}^+ X$. In particular, $\tau_{\mathcal{C}}^- \tau_{\mathcal{C}}^+ X \cong X$ and $\theta_{\mathcal{C}}^- \tau_{\mathcal{C}}^+ X \cong \theta_{\mathcal{C}}^+ X$. The dual statement holds if $\tau_{\mathcal{C}}^- X \neq 0$.*
- (3) *The maps $\tau_{\mathcal{C}}^+$ and $\tau_{\mathcal{C}}^-$ are mutually inverse bijections between $\text{ind } \mathcal{C} \setminus \text{ind}_1^+ \mathcal{C}$ and $\text{ind } \mathcal{C} \setminus \text{ind}_1^- \mathcal{C}$.*

Example 2.8. The category $\text{mod } A$ is a τ -category: its right and left τ -sequences are the almost split sequences with the zeros at both ends omitted, together with $0 \rightarrow \text{rad } P \rightarrow P$ at projectives and $I \rightarrow I/\text{soc } I \rightarrow 0$ at injectives. The almost split factorization properties give the two exactness conditions in Definition 2.6; the ordinary kernels and cokernels give the required minimal weak kernels and cokernels. Thus $\tau_{\text{mod } A}^+ = \tau_A$, $\tau_{\text{mod } A}^- = \tau_A^-$, $\theta_{\text{mod } A}^+ = \theta_A$, and $\theta_{\text{mod } A}^- = \theta_A^-$ in the notation of Definition 2.1, and the τ -projectives and τ -injectives are the projectives and injectives. Each ν_X^+ is monic, and each ν_X^- is epic. These facts are also noted in [15, Section 1.3].

As for modules [3, VII, Proposition 1.3], the middle terms of τ -sequences record the irreducible maps; for τ -categories this is stated in [16, Section 9.1].

Lemma 2.9. *Let $\mathcal{C}$ be a Hom-finite k -linear τ -category, and let X be an indecomposable. For every indecomposable Y ,*

$$[\theta_{\mathcal{C}}^+ X : Y]_{\oplus} = \dim_k \text{Irr}_{\mathcal{C}}(Y, X), \quad [\theta_{\mathcal{C}}^- X : Y]_{\oplus} = \dim_k \text{Irr}_{\mathcal{C}}(X, Y).$$

Proof. We prove the first formula; the second follows by applying it to the opposite category. By (2.3), $P_{\mu_X^+}$ maps $P_{\theta_{\mathcal{C}}^+ X}$ onto $\text{rad}_{\mathcal{C}}(-, X)$, and its kernel is the image of $P_{\nu_X^+}$, which lies in $\text{rad}_{\mathcal{C}}(-, \theta_{\mathcal{C}}^+ X)$ because ν_X^+ is radical. The image of $\text{rad}_{\mathcal{C}}(-, \theta_{\mathcal{C}}^+ X)$ is $\text{rad}_{\mathcal{C}}^2(-, X)$: it lies there because μ_X^+ is radical, and conversely a composite ab of radical maps $b : Y \rightarrow W$ and $a : W \rightarrow X$ has the form $\mu_X^+(cb)$, where $a = \mu_X^+c$. Hence $P_{\mu_X^+}$ induces an isomorphism

$$P_{\theta_{\mathcal{C}}^+ X} / \text{rad}_{\mathcal{C}}(-, \theta_{\mathcal{C}}^+ X) \cong \text{rad}_{\mathcal{C}}(-, X) / \text{rad}_{\mathcal{C}}^2(-, X).$$

The left side is the direct sum of the functors S_V over the indecomposable summands V of $\theta_{\mathcal{C}}^+ X$, with multiplicity. Evaluate at Y . Since $S_V(Y) = 0$ for $V \not\cong Y$ and $S_Y(Y) = \text{End}_{\mathcal{C}}(Y) / \text{rad}_{\mathcal{C}}(Y, Y) = k$, the left side has dimension $[\theta_{\mathcal{C}}^+ X : Y]_{\oplus}$, and the right side is $\text{Irr}_{\mathcal{C}}(Y, X)$. $\square$

We will repeatedly express radical maps as sums of composites of irreducible maps. For module categories this is [3, V, Theorem 7.8]; see also [1, IV, Corollary 5.6].

Lemma 2.10. *Let $\mathcal{C}$ be a Hom-finite Krull–Schmidt k -linear category with finitely many indecomposable isomorphism classes. Choose maps representing a basis of each $\text{Irr}_{\mathcal{C}}(X, Y)$. Then the radical of $\mathcal{C}$ is nilpotent, and every radical map between indecomposables is a linear combination of nonempty paths in the chosen maps.*

Proof. Put $M = \bigoplus_{X \in \text{ind } \mathcal{C}} X$ and $\Lambda = \text{End}_{\mathcal{C}}(M)$. The spaces $\text{rad}_{\mathcal{C}}(X, Y)$ are the corresponding matrix blocks of $\text{rad } \Lambda$. Since Λ is finite-dimensional, choose N with $(\text{rad } \Lambda)^N = 0$; then $\text{rad}_{\mathcal{C}}^N = 0$. Modulo $\text{rad}_{\mathcal{C}}^2$, every radical map is a linear combination of the chosen representatives. Repeating this expansion in each factor expresses it modulo $\text{rad}_{\mathcal{C}}^{n+1}$ as a sum of paths of lengths at most n . Taking $n = N - 1$ proves the claim. $\square$

2.2. Magnitude in a strict τ -category. The following definition extends magnitude from module categories to Krull–Schmidt categories. Proposition 2.14 below computes it for ideal quotients and categories of poset representations; we first describe the categories to which it applies.

Definition 2.11. Let $\mathcal{C}$ be a Hom-finite Krull–Schmidt k -linear category with finitely many indecomposable isomorphism classes. Its *Hom matrix* and *magnitude* are

$$H_{\mathcal{C}} = (\dim_k \mathcal{C}(X, Y))_{X, Y \in \text{ind } \mathcal{C}}, \quad \chi(\mathcal{C}) = \mathbf{1}^T H_{\mathcal{C}}^{-1} \mathbf{1},$$

the latter whenever $H_{\mathcal{C}}$ is invertible.

Definition 2.12. Following [15, Section 1.3], a τ -category $\mathcal{C}$ is *strict* if ν_X^+ is monic and ν_X^- is epic for every indecomposable X .

By Example 2.8, $\text{mod } A$ is a strict τ -category. In a strict τ -category, the presentations (2.3) become resolutions; cf. [16, Section 2.1] and the proof of [15, Theorem 4.3].

Lemma 2.13. *Let $\mathcal{C}$ be a strict τ -category and X an indecomposable. Then the sequences*

$$\begin{aligned} 0 \longrightarrow P_{\tau_{\mathcal{C}}^+ X} \longrightarrow P_{\theta_{\mathcal{C}}^+ X} \longrightarrow P_X \longrightarrow S_X \longrightarrow 0, \\ 0 \longrightarrow P^{\tau_{\mathcal{C}}^- X} \longrightarrow P^{\theta_{\mathcal{C}}^- X} \longrightarrow P^X \longrightarrow S^X \longrightarrow 0, \end{aligned}$$

induced by the right and left τ -sequences at X , are exact.

Proof. By (2.3), the first sequence is exact except possibly at $P_{\tau_{\mathcal{C}}^+ X}$, and there it is exact because ν_X^+ is monic. The second sequence is dual. $\square$

Proposition 2.14. *Let $\mathcal{C}$ be a Hom-finite strict τ -category with finitely many indecomposable isomorphism classes. Let $a(\mathcal{C}) = \sum_{X, Y} \dim_k \text{Irr}_{\mathcal{C}}(X, Y)$. Then $H_{\mathcal{C}}$ is invertible and*

$$\chi(\mathcal{C}) = 2|\text{ind } \mathcal{C}| - a(\mathcal{C}) - |\text{ind}_1^+ \mathcal{C}|. \quad (2.4)$$

Proof. The argument is that of [6, Lemma 1.9], which applies [8, Theorem 1.1] to these resolutions; the inverse of $H_{\mathcal{C}}$ below also appears in the proof of [15, Theorem 4.3]. Evaluate the first exact sequence of Lemma 2.13 at each indecomposable Y . Let M be the matrix whose X -column is the multiplicity vector of $[X] - [\theta_{\mathcal{C}}^+ X] + [\tau_{\mathcal{C}}^+ X]$, where $[Z]$ records the multiplicities of the indecomposable summands of Z . The row of $H_{\mathcal{C}}$ indexed by Y lists the dimensions $\dim_k \mathcal{C}(Y, Z)$, and $\dim_k S_X(Y)$ is 1 if $X = Y$ and 0 otherwise, so the evaluated sequence gives

$$(H_{\mathcal{C}} M)_{Y, X} = \dim_k \mathcal{C}(Y, X) - \dim_k \mathcal{C}(Y, \theta_{\mathcal{C}}^+ X) + \dim_k \mathcal{C}(Y, \tau_{\mathcal{C}}^+ X) = \mathbf{1}_{\{X=Y\}}.$$

Thus M is a right inverse of $H_{\mathcal{C}}$, and therefore its inverse. The total multiplicity of the middle terms is $a(\mathcal{C})$, by Lemma 2.9. There are $|\text{ind } \mathcal{C}| - |\text{ind}_1^+ \mathcal{C}|$ nonzero translates, each indecomposable. Summing all entries gives (2.4). $\square$

In other words, $\chi(\mathcal{C})$ is the number of vertices of the irreducible quiver of $\mathcal{C}$, minus the number of its arrows, plus the number of vertices with a nonzero translate.

2.3. Magnitude of module categories. Since the τ -projectives of $\text{mod } A$ correspond to the simple modules, Proposition 2.14 recovers the formula (1.1) of Børve, Horiatakis, and Kalck. We record it together with the surplus studied in the rest of the paper. Fix a representation-finite algebra A .

Definition 2.15. Write $a(A) = a(\text{mod } A)$, and put $\sigma(A) = \chi(\text{mod } A) - |A|$. We call $\sigma(A)$ the *magnitude surplus*.

Lemma 2.16 ([6, Lemma 1.9]). *The matrix $H_A = (\dim_k \text{Hom}_A(X, Y))_{X, Y \in \text{ind } A}$ is invertible, and*

$$\chi(\text{mod } A) = 2|\text{ind } A| - |A| - a(A).$$

Consequently,

$$\sigma(A) = 2(|\text{ind } A| - |A|) - a(A). \quad (2.5)$$

Remark 2.17. Børve, Horiatakis, and Kalck also give a refinement of (1.1), in which the meshes of Γ_A are counted according to the number of indecomposable middle terms of the corresponding almost split sequences [6, Lemma 1.13].

The following lemma reduces statements to basic algebras in Sections 4–6 and computes the surplus of products of interval algebras in Section 6; cf. [6, Remark 1.7(2)].

Lemma 2.18. *Magnitude, $|A|$, and $\sigma(A)$ are Morita invariant and additive on products of algebras.*

Proof. A k -linear Morita equivalence preserves simple classes, indecomposable classes, and Hom dimensions. For a product of algebras, the Hom matrix is block diagonal, so its inverse is block diagonal and its entry sum is the sum for the factors. The simple classes are the disjoint union of those of the factors. Subtracting their numbers proves the assertion for σ . $\square$

3. MAGNITUDE OF SERRE SUBCATEGORIES AND IDEAL QUOTIENTS

An ideal quotient removes objects by setting their identity maps equal to zero. We first describe what this operation does to indecomposables and irreducible maps in an additive category. For a Serre subcategory of a module category, we then show that the ideal quotient is a strict τ -category, so that Proposition 2.14 computes its magnitude, and compare the almost split sequences of the subcategory with those of the module category. Theorem 3.12 relates $\text{mod } A$, the subcategory, and the ideal quotient through two counts c and r from this comparison and a correction term δ . Theorem 3.15 then gives a sufficient condition for the inequality $r \leq c$.

3.1. Indecomposables and irreducible maps in ideal quotients. We recall ideal quotients and describe their indecomposables and irreducible maps.

Definition 3.1. Let $\mathcal{C}$ be an additive category and $\mathcal{U}$ a full additive subcategory closed under direct summands. The *ideal quotient* $\mathcal{C}/[\mathcal{U}]$ has the same objects as $\mathcal{C}$, and

$$(\mathcal{C}/[\mathcal{U}])(X, Y) = \mathcal{C}(X, Y)/[\mathcal{U}](X, Y),$$

where $[\mathcal{U}](X, Y)$ consists of the maps factoring through an object of $\mathcal{U}$. Write π for the quotient functor.

Proposition 3.2. *Suppose that $\mathcal{C}$ is Hom-finite, k -linear, and Krull–Schmidt. Then $\mathcal{C}/[\mathcal{U}]$ is Krull–Schmidt. Its indecomposable isomorphism classes are indexed by $\text{ind } \mathcal{C} \setminus \text{ind } \mathcal{U}$. For representatives X, Y in this complement, π induces*

$$\text{Irr}_{\mathcal{C}}(X, Y) \cong \text{Irr}_{\mathcal{C}/[\mathcal{U}]}(\pi X, \pi Y).$$

Proof. For an indecomposable $X \notin \mathcal{U}$, the ideal $[\mathcal{U}](X, X)$ is proper: otherwise X would be a summand of an object of $\mathcal{U}$. It is therefore contained in the radical of the local ring $\text{End}_{\mathcal{C}}(X)$. Its quotient is local, so πX remains indecomposable. If $\pi X \cong \pi Y$, lift inverse maps to $f : X \rightarrow Y$ and $g : Y \rightarrow X$. Both gf and fg are identities modulo radical ideals, hence invertible. Thus $X \cong Y$. Every object is a finite sum of these objects with local endomorphism rings, proving the Krull–Schmidt assertion.

For the irreducible maps, Iyama shows for an ideal quotient of an additive category that $\text{rad}_{\mathcal{C}/[\mathcal{U}]}^n = (\text{rad}_{\mathcal{C}}^n + [\mathcal{U}]) / [\mathcal{U}]$ for $n \geq 1$ [17, Section 1.3(5)]. Let $X, Y \in \text{ind } \mathcal{C} \setminus \text{ind } \mathcal{U}$. A map $X \rightarrow Y$ which factors through an indecomposable of $\mathcal{U}$ is a composite of two radical maps, so $[\mathcal{U}](X, Y) \subseteq \text{rad}_{\mathcal{C}}^2(X, Y)$. The case $n = 1, 2$ of Iyama's formula therefore gives the asserted isomorphism. $\square$

3.2. Ideal quotients by Serre subcategories. We now specialize to a Serre subcategory of a module category.

Setting 3.3. Let A be a finite-dimensional algebra, and e an idempotent of A . Put

$$B = A/AeA, \quad \mathcal{D} = (\text{mod } A)/[\text{mod } B],$$

where $\text{mod } B$ is regarded as the full subcategory of $\text{mod } A$ of modules annihilated by AeA . Since $ZAeA = ZeA$ for a module Z , these are the modules Z with $Ze = 0$. Since evaluation at e is exact, $\text{mod } B$ is a Serre subcategory, that is, closed under submodules, quotients, and extensions. Write $i : \text{mod } B \hookrightarrow \text{mod } A$ for the inclusion and $\pi : \text{mod } A \rightarrow \mathcal{D}$ for the ideal quotient. The inclusion has the right and left adjoints

$$i^!Z = \{z \in Z : zAeA = 0\} \cong \text{Hom}_A(B, Z), \quad i^*Z = Z/ZeA \cong Z \otimes_A B,$$

the largest submodule and the largest quotient of Z in $\text{mod } B$. By extension closure, $Z/i^!Z$ has no nonzero submodule in $\text{mod } B$, and dually $\ker(Z \rightarrow i^*Z)$ has no nonzero quotient in $\text{mod } B$.

Every Serre subcategory of $\text{mod } A$ arises in this way: choose e as the sum of primitive idempotents corresponding to the simple modules outside it. For a module map $f : X \rightarrow Y$, write $f_e : Xe \rightarrow Ye$ for its restriction. Ringel and Vossieck show that a map of A -modules factors through $\text{mod } B$ exactly when its restriction f_e is zero [21, Section 10, Lemma 1]; this is the faithfulness in the following lemma.

Lemma 3.4. *Evaluation at e induces a faithful functor $F : \mathcal{D} \rightarrow \text{mod } Ae$, $\pi X \mapsto Xe$. In particular, a map in $\mathcal{D}$ is monic, respectively epic, if its image under F is.*

Proof. A map $f : X \rightarrow Y$ with $f_e = 0$ kills XeA , so it factors through $X/XeA \in \text{mod } B$. Conversely, a map factoring through $\text{mod } B$ has zero evaluation at e . Faithful functors reflect monomorphisms and epimorphisms. $\square$

Remark 3.5. The ideal quotient $\mathcal{D}$ differs from the abelian Serre quotient $(\text{mod } A)/(\text{mod } B)$, in which every map with kernel and cokernel in $\text{mod } B$ becomes invertible. Let $A = (k(1 \rightarrow 2))^{\text{op}}$ and $e = e_2$, so that $\text{mod } B = \text{add } S_1$. The inclusion $S_2 \rightarrow P_1$ has cokernel S_1 , so it becomes invertible in $(\text{mod } A)/(\text{mod } B)$. Under F it becomes the identity of k , so it is monic and epic in $\mathcal{D}$. It is not invertible in $\mathcal{D}$, because $\text{Hom}_A(P_1, S_2) = 0$.

To find the τ -sequences of $\mathcal{D}$, we use Iyama's description of τ -sequences in an ideal quotient.

Theorem 3.6 ([17, Section 1.4(2)]). *Let $\mathcal{C}$ be a τ -category and $\mathcal{U}$ a full additive subcategory closed under direct summands, with quotient functor $\pi : \mathcal{C} \rightarrow \mathcal{C}/[\mathcal{U}]$. Then $\mathcal{C}/[\mathcal{U}]$ is a τ -category. Let $X \in \text{ind } \mathcal{C} \setminus \text{ind } \mathcal{U}$. If $\pi(\theta_{\mathcal{C}}^+ X) \neq 0$, the right τ -sequence ending at πX is*

$$\pi(\tau_{\mathcal{C}}^+ X) \xrightarrow{\pi\nu_X^+} \pi(\theta_{\mathcal{C}}^+ X) \xrightarrow{\pi\mu_X^+} \pi X;$$

otherwise it is $0 \rightarrow 0 \rightarrow \pi X$.

Applied to the opposite categories, the theorem gives the dual description of left τ -sequences.

Proposition 3.7. *The category $\mathcal{D}$ in Setting 3.3 is a strict τ -category. For $X \in \text{ind } A \setminus \text{ind } B$, the right τ -sequence ending at πX and the left τ -sequence starting at πX are*

$$\pi(\tau_A X) \longrightarrow \pi(\theta_A X) \longrightarrow \pi X, \quad \pi X \longrightarrow \pi(\theta_A^- X) \longrightarrow \pi(\tau_A^- X).$$

In particular, $\tau_{\mathcal{D}}^+(\pi X) \cong \pi(\tau_A X)$ and $\tau_{\mathcal{D}}^-(\pi X) \cong \pi(\tau_A^- X)$.

Proof. Apply Theorem 3.6 to the τ -category $\text{mod } A$ of Example 2.8 and $\mathcal{U} = \text{mod } B$. If X is nonprojective, exactness of $(-)_e$ gives $(\theta_A X)e \rightarrow Xe \neq 0$; hence $\pi(\theta_A X) \neq 0$, and the right τ -sequence of πX is the image of the almost split sequence ending at X . If X is projective, then $\tau_A X = 0$ and $\theta_A X = \text{rad } X$, and both cases of the theorem give the sequence $0 \rightarrow \pi(\text{rad } X) \rightarrow \pi X$. This proves the description of right τ -sequences; that of left τ -sequences is dual. The first maps $\pi\nu_X^+$ of these right τ -sequences become monic under F , by exactness of $(-)_e$, and the last maps $\pi\nu_X^-$ of the left τ -sequences become epic. Lemma 3.4 proves strictness. $\square$

When A is representation-finite, the strictness of $\mathcal{D}$ is also a case of [17, Theorem 6.1], since $i^!$ and i^* have monic counit and epic unit.

3.3. Restricting almost split sequences to Serre subcategories. By Proposition 3.7, the translate $\tau_{\mathcal{D}}^+(\pi X)$ is $\pi(\tau_A X)$, which is zero exactly when $\tau_A X$ lies in $\text{mod } B$. The translate of a B -module in $\text{mod } B$ is obtained differently: by the following result of Hoshino, it is the largest submodule of its translate in $\text{mod } A$ that belongs to $\text{mod } B$.

Proposition 3.8 ([14, Section 1, Lemmas 2(2) and 3(2)]). *Let Y be a nonprojective indecomposable of $\text{mod } B$, and write its almost split sequence in $\text{mod } A$ as $0 \rightarrow \tau_A Y \rightarrow \theta_A Y \rightarrow Y \rightarrow 0$. Then*

$$0 \longrightarrow i^!(\tau_A Y) \longrightarrow i^!(\theta_A Y) \longrightarrow Y \longrightarrow 0 \quad (3.1)$$

is its almost split sequence in $\text{mod } B$. In particular, $\tau_B Y \cong i^!(\tau_A Y)$. Dually, applying i^ to the ambient almost split sequence starting at a noninjective $X \in \text{ind } B$ gives its sequence in $\text{mod } B$, with $\tau_B^- X \cong i^*(\tau_A^- X)$.*

Proof. The lemmas of Hoshino concern a torsion pair in a module category, and they are essentially due to Auslander and Smalø [4]. Lemma 2(2) of [14] states the following. Let $\mathcal{T}$ be a torsion class with torsion radical t , and let $Y \in \mathcal{T}$ be indecomposable and not Ext-projective in $\mathcal{T}$, that is, $\text{Ext}_A^1(Y, \mathcal{T}) \neq 0$. Then applying t to the almost split sequence of $\text{mod } A$ ending at Y gives an almost split sequence in $\mathcal{T}$. Lemma 3(2) is the dual statement for a torsion-free class and its torsion-free quotient functor.

The subcategory $\text{mod } B$ is closed under quotients and extensions, so it is a torsion class, with torsion radical $i^!$. It is closed under submodules and extensions, so it is also the torsion-free class of another torsion pair, with torsion-free quotient functor i^* . Since $\text{mod } B$ is extension-closed, $\text{Ext}_B^1 = \text{Ext}_A^1$ on $\text{mod } B$, so its Ext-projective and Ext-injective objects are the projective and injective B -modules. The two lemmas therefore give the two assertions. $\square$

Example 3.9. Let $A = (k(1 \rightarrow 2 \leftarrow 3))^{\text{op}}$ and $e = e_3$, so that $\text{mod } B$ consists of the representations vanishing at 3. The ambient sequence ending at S_1 is

$$0 \longrightarrow P_3 \longrightarrow I_2 \longrightarrow S_1 \longrightarrow 0.$$

Applying $i^!$ gives

$$0 \longrightarrow S_2 \longrightarrow P_1 \longrightarrow S_1 \longrightarrow 0,$$

the almost split sequence in $\text{mod } B \simeq \text{mod}((k(1 \rightarrow 2))^{\text{op}})$. In particular $\tau_A S_1 = P_3$, whereas $\tau_B S_1 = S_2$. The map $P_1 \rightarrow S_1$ is irreducible in $\text{mod } B$, but factors through I_2 in $\text{mod } A$.

3.4. Magnitudes of a Serre subcategory and its ideal quotient. We now compare the irreducible maps and the almost split sequences of $\text{mod } B$ with those of $\text{mod } A$, and use the comparison to relate the magnitudes of $\text{mod } A$, $\text{mod } B$, and $\mathcal{D}$. Assume from now on in this section that B is representation-finite.

Definition 3.10. For $V, W \in \text{ind } B$, put

$$c_{VW} = \dim_k \text{Irr}_B(V, W) - \dim_k \text{Irr}_A(V, W), \quad c = \sum_{V, W \in \text{ind } B} c_{VW}.$$

Let r be the number of isomorphism classes of almost split sequences in $\text{mod } B$ that are not almost split in $\text{mod } A$.

Lemma 3.11. *The identity on $\text{rad}_B(V, W) = \text{rad}_A(V, W)$ induces a surjection $\text{Irr}_B(V, W) \twoheadrightarrow \text{Irr}_A(V, W)$. Its kernel is $\text{rad}_A^2(V, W) / \text{rad}_B^2(V, W)$ and has dimension $c_{VW} \geq 0$.*

Proof. Fullness gives equality of the radicals. Every factorization through $\mathbf{mod} B$ is also one through $\mathbf{mod} A$, so $\text{rad}_B^2(V, W) \subseteq \text{rad}_A^2(V, W)$. Taking the two quotients proves the assertion. $\square$

Thus Γ_B has c_{VW} more arrows from V to W than Γ_A has.

The following identity expresses $\chi(\mathbf{mod} A) - \chi(\mathbf{mod} B) - \chi(\mathcal{D})$ through c , r , and the middle terms of the almost split sequences of $\mathbf{mod} A$ with exactly one end term in $\mathbf{mod} B$.

Theorem 3.12. *Assume A is representation-finite. For each almost split sequence of $\mathbf{mod} A$ with exactly one end term in $\mathbf{mod} B$, let m_{out} be the number of its indecomposable middle summands outside $\mathbf{mod} B$, counting multiplicities, and put*

$$\delta = \sum_{\text{such sequences}} (m_{\text{out}} - 1).$$

Then $\delta \geq 0$, and

$$\chi(\mathbf{mod} A) - \chi(\mathbf{mod} B) - \chi(\mathcal{D}) = c - r - \delta. \quad (3.2)$$

Proof. Let $0 \rightarrow \tau_A Y \rightarrow \theta_A Y \rightarrow Y \rightarrow 0$ be a sequence in the sum. Since $Ye \neq 0$ or $(\tau_A Y)e \neq 0$, applying the exact functor $(-)e$ gives $(\theta_A Y)e \neq 0$. Thus $\theta_A Y$ has an indecomposable summand outside $\mathbf{mod} B$, and $m_{\text{out}} \geq 1$. By Proposition 3.2, $|\text{ind } A| = |\text{ind } \mathcal{D}| + |\text{ind } B|$. Let t be the number of sequences in the sum.

We count the $|\text{ind } A| - |A|$ almost split sequences of $\mathbf{mod} A$ by their end terms. A sequence with both end terms in $\mathbf{mod} B$ lies in $\mathbf{mod} B$, by extension closure, and remains almost split there. These are the almost split sequences of $\mathbf{mod} B$ not counted by r , and there are $|\text{ind } B| - |B| - r$ of them. By Proposition 3.7, the sequences with both end terms outside $\mathbf{mod} B$ correspond to the objects of $\mathcal{D}$ with nonzero translate, and there are $|\text{ind } \mathcal{D}| - |\text{ind}_1^+ \mathcal{D}|$ of them. Hence

$$|\text{ind } A| - |A| = (|\text{ind } \mathcal{D}| - |\text{ind}_1^+ \mathcal{D}|) + (|\text{ind } B| - |B| - r) + t,$$

and substituting $|\text{ind } A| = |\text{ind } \mathcal{D}| + |\text{ind } B|$ gives

$$t = |\text{ind}_1^+ \mathcal{D}| - (|A| - |B|) + r. \quad (3.3)$$

Let z be the number of arrows of Γ_A with exactly one endpoint in $\mathbf{mod} B$, counted with multiplicity. Let $V \rightarrow W$ be such an arrow with $V \in \mathbf{mod} B$. An arrow out of an injective A -module in $\mathbf{mod} B$ is epic, so its target also lies in $\mathbf{mod} B$; hence V is not injective. The almost split sequence starting at V has W as a middle summand, so its last term is not in $\mathbf{mod} B$, by extension closure. Thus this sequence is counted by t , and W is one of the middle summands outside $\mathbf{mod} B$ counted by m_{out} for this sequence. Dually, such an arrow $W \rightarrow V$ with $V \in \mathbf{mod} B$ ends at a nonprojective V and gives a middle summand outside $\mathbf{mod} B$ of the sequence ending at V . Conversely, every middle summand outside $\mathbf{mod} B$ of a sequence counted by t gives an arrow to or from its end term in $\mathbf{mod} B$. Since such a sequence has only one end term in $\mathbf{mod} B$, each such arrow arises from exactly one sequence, the one starting or ending at its endpoint in $\mathbf{mod} B$. We therefore obtain a bijection between the arrows of Γ_A with exactly one endpoint in $\mathbf{mod} B$ and the pairs of a sequence counted by t and one of its middle summands outside $\mathbf{mod} B$. By Proposition 2.2 it respects multiplicities, so $z = \sum m_{\text{out}}$, and hence

$$\delta = z - t. \quad (3.4)$$

The arrows of Γ_A between objects outside $\mathbf{mod} B$ are the arrows of $\mathcal{D}$ by Proposition 3.2; those between objects of $\mathbf{mod} B$ number $a(B) - c$ by Definition 3.10; the remaining z arrows have exactly one endpoint in $\mathbf{mod} B$. Hence

$$a(A) = a(\mathcal{D}) + z + a(B) - c. \quad (3.5)$$

Lemma 2.16 and Proposition 2.14 give

$$\begin{aligned} \chi(\mathbf{mod} A) &= 2|\text{ind } A| - |A| - a(A), & \chi(\mathbf{mod} B) &= 2|\text{ind } B| - |B| - a(B), \\ \chi(\mathcal{D}) &= 2|\text{ind } \mathcal{D}| - a(\mathcal{D}) - |\text{ind}_1^+ \mathcal{D}|. \end{aligned} \quad (3.6)$$

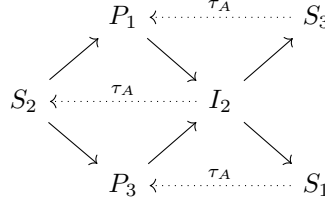
Subtract the last two formulas in (3.6) from the first, and substitute $|\text{ind } A| = |\text{ind } \mathcal{D}| + |\text{ind } B|$ and (3.5). Then (3.3) and (3.4) give

$$\chi(\text{mod } A) - \chi(\text{mod } B) - \chi(\mathcal{D}) = |\text{ind}_1^+ \mathcal{D}| - (|A| - |B|) + c - z = (t - r) + c - z = c - r - \delta.$$

□

For a primitive idempotent of a representation-directed algebra, we show in Proposition 4.19 that $\delta = 0$, using Proposition 4.12. We continue Example 3.9 and compute every term of Theorem 3.12.

Example 3.13. Keep $A = (k(1 \rightarrow 2 \leftarrow 3))^{\text{op}}$ and $e = e_3$. The Auslander–Reiten quiver of A is



where $S_2 = P_2$, $S_1 = I_1$, and $S_3 = I_3$. It has six vertices and six arrows, so $\chi(\text{mod } A) = 12 - 3 - 6 = 3$ by Lemma 2.16. The category $\text{mod } B$ has the three indecomposables S_2 , P_1 , S_1 and the two arrows $S_2 \rightarrow P_1 \rightarrow S_1$, so $\chi(\text{mod } B) = 6 - 2 - 2 = 2$.

The indecomposables of $\mathcal{D}$ are P_3 , I_2 , and S_3 , with the two arrows $P_3 \rightarrow I_2 \rightarrow S_3$. By Proposition 3.7, $\tau_{\mathcal{D}}^+ I_2$ and $\tau_{\mathcal{D}}^+ S_3$ are the images of S_2 and P_1 , which are zero in $\mathcal{D}$, and P_3 is projective. Thus all three objects are τ -projective, although I_2 and S_3 are not projective over A , and Proposition 2.14 gives $\chi(\mathcal{D}) = 6 - 2 - 3 = 1$.

Of the two arrows of Γ_B , the arrow $S_2 \rightarrow P_1$ is an arrow of Γ_A , and $P_1 \rightarrow S_1$ is not; hence $c = 1$. The only almost split sequence of $\text{mod } B$ is not almost split in $\text{mod } A$, so $r = 1$. The almost split sequences of $\text{mod } A$ with exactly one end term in $\text{mod } B$ end at I_2 , S_1 , and S_3 . Each has exactly one middle summand outside $\text{mod } B$, namely P_3 , I_2 , and I_2 , so $\delta = 0$. Both sides of (3.2) are zero.

3.5. Almost split sequences not preserved by the inclusion. In the identity (3.2), the difference $c - r$ is nonnegative under a condition on the extensions between $\text{mod } B$ and the simple modules outside it. Let E be a direct sum of one representative of each simple module outside $\text{mod } B$. Define functors on $\text{mod } B$ by

$$\mathbf{H}_E(W) = \text{Ext}_A^1(E, W), \quad \mathbf{G}_E(W) = \text{Ext}_A^1(W, E).$$

Lemma 3.14. Let $P_E \twoheadrightarrow E$ and $E \hookrightarrow I_E$ be the projective cover and injective envelope, and let $\xi : 0 \rightarrow X \rightarrow \theta_B Y \rightarrow Y \rightarrow 0$ be an almost split sequence in $\text{mod } B$. Put $V = \tau_A Y$ and $Q = V/i^! V$.

(1) $\mathbf{H}_E \cong \text{Hom}_B(i^* \text{rad } P_E, -)$ and $\mathbf{G}_E \cong \text{Hom}_B(-, i^!(I_E/E))$.

(2) $\text{Ext}_A^1(E, V) = 0$, and there is an exact sequence

$$0 \longrightarrow D_k \mathbf{G}_E(Y) \longrightarrow \text{Hom}_A(E, Q) \longrightarrow \mathbf{H}_E(X) \longrightarrow 0. \quad (3.7)$$

(3) The sequence ξ is not almost split in $\text{mod } A$ if and only if $\mathbf{H}_E(X) \neq 0$ or $\mathbf{G}_E(Y) \neq 0$.

Proof. (1) For $W \in \text{mod } B$, the top of P_E and socle of I_E give $\text{Hom}_A(P_E, W) = 0 = \text{Hom}_A(W, I_E)$. Applying $\text{Hom}_A(-, W)$ to $0 \rightarrow \text{rad } P_E \rightarrow P_E \rightarrow E \rightarrow 0$ gives $\text{Ext}_A^1(E, W) \cong \text{Hom}_A(\text{rad } P_E, W) \cong \text{Hom}_B(i^* \text{rad } P_E, W)$, by the adjunction for i^* . Dually, applying $\text{Hom}_A(W, -)$ to $0 \rightarrow E \rightarrow I_E \rightarrow I_E/E \rightarrow 0$ and using the adjunction for $i^!$ gives the second representation.

(2) By Proposition 3.8, identify $X = i^! V$. The first isomorphism in (2.2), with $M = E$ and $N = V$, gives

$$\text{Ext}_A^1(E, V) = D_k \underline{\text{Hom}}_A(Y, E) = 0,$$

since $\text{Hom}_A(Y, E) = 0$ for $Y \in \text{mod } B$. The second, with $M = Y$ and $N = E$, gives $D_k \mathbf{G}_E(Y) = \overline{\text{Hom}}_A(E, V) = \text{Hom}_A(E, V)$. For the equality $\overline{\text{Hom}}_A(E, V) = \text{Hom}_A(E, V)$, let a nonzero map $S \rightarrow V$ from a simple summand S of E factor through an injective module. It then factors through the injective envelope $S \hookrightarrow I(S)$. The resulting map $I(S) \rightarrow V$ is monic on the essential simple socle S , hence monic and split. This contradicts the fact that the indecomposable V is not injective. Applying $\text{Hom}_A(E, -)$ to $0 \rightarrow X \rightarrow V \rightarrow Q \rightarrow 0$, and using $\text{Hom}_A(E, X) = 0$, proves (3.7).

(3) The sequence ξ is almost split in $\mathbf{mod} A$ precisely when $V \in \mathbf{mod} B$, equivalently when $Q = 0$. Indeed, if ξ is almost split in $\mathbf{mod} A$, then $V \cong X \in \mathbf{mod} B$. Conversely, if $V \in \mathbf{mod} B$, then the ambient sequence lies in $\mathbf{mod} B$ by extension closure, and it coincides with ξ by Proposition 3.8. Every simple subobject of a nonzero Q is outside $\mathbf{mod} B$, by Setting 3.3. Thus $Q \neq 0$ is equivalent to $\mathrm{Hom}_A(E, Q) \neq 0$, and (2) proves the criterion. $\square$

Theorem 3.15. *Suppose that*

$$\mathrm{H}_E(W) = 0 \quad \text{or} \quad \mathrm{G}_E(W) = 0 \quad \text{for every } W \in \mathrm{ind} B. \quad (3.8)$$

Then $r \leq c$. More precisely, the almost split sequences of $\mathbf{mod} B$ that are not almost split in $\mathbf{mod} A$ can be assigned distinct ordered pairs (V, W) with $c_{VW} > 0$.

Proof. An irreducible map $V \rightarrow W$ in the module category $\mathbf{mod} B$ is monic or epic. In the monic case, representability of H_E (Lemma 3.14(1)) gives an injection $\mathrm{H}_E(V) \hookrightarrow \mathrm{H}_E(W)$; in the epic case, representability of G_E gives an injection $\mathrm{G}_E(W) \hookrightarrow \mathrm{G}_E(V)$. Hence, if $\mathrm{Irr}_B(V, W) \neq 0$, $\mathrm{H}_E(V) \neq 0$, and $\mathrm{G}_E(W) \neq 0$, then $\mathrm{H}_E(W) \neq 0 \neq \mathrm{G}_E(W)$ in the monic case and $\mathrm{H}_E(V) \neq 0 \neq \mathrm{G}_E(V)$ in the epic case. Thus (3.8) implies

$$\mathrm{Irr}_B(V, W) \neq 0 \implies \mathrm{H}_E(V) = 0 \text{ or } \mathrm{G}_E(W) = 0. \quad (3.9)$$

Consider an almost split sequence $0 \rightarrow X \rightarrow \theta_B Y \rightarrow Y \rightarrow 0$ of $\mathbf{mod} B$ that is not almost split in $\mathbf{mod} A$. Since $\mathrm{Hom}_A(E, Y) = 0$, its long exact sequence gives $\mathrm{H}_E(X) \hookrightarrow \mathrm{H}_E(\theta_B Y)$. If $\mathrm{H}_E(X) \neq 0$, some middle summand W has $\mathrm{H}_E(W) \neq 0$, and its arrow to Y , together with (3.9), forces $\mathrm{G}_E(Y) = 0$. By Lemma 3.14(3), at least one of $\mathrm{H}_E(X)$ and $\mathrm{G}_E(Y)$ is nonzero, so exactly one is. We assign to the sequence an ordered pair of indecomposable B -modules, in two cases.

Case 1: $\mathrm{H}_E(X) \neq 0$. Write the ambient middle term ending at Y as $\theta_A Y = M_{\mathrm{in}} \oplus M_{\mathrm{out}}$, where the summands of M_{in} lie in $\mathbf{mod} B$ and none of those of M_{out} do. Proposition 3.8 gives

$$\theta_B Y = M_{\mathrm{in}} \oplus i^! M_{\mathrm{out}}. \quad (3.10)$$

Let $W \in \mathrm{ind} B$. By Proposition 2.2 in $\mathbf{mod} B$ and in $\mathbf{mod} A$, $[\theta_B Y : W]_{\oplus} = \dim_k \mathrm{Irr}_B(W, Y)$ and $[M_{\mathrm{in}} : W]_{\oplus} = [\theta_A Y : W]_{\oplus} = \dim_k \mathrm{Irr}_A(W, Y)$. Subtracting and using (3.10) gives

$$[i^! M_{\mathrm{out}} : W]_{\oplus} = c_{WY}. \quad (3.11)$$

The component $X \rightarrow M_{\mathrm{in}}$ of the first map of ξ is the restriction of $\nu_Y^+ : \tau_A Y \rightarrow \theta_A Y$ to $X = i^!(\tau_A Y)$, followed by the projection to M_{in} . Hence the map $\mathrm{H}_E(X) \rightarrow \mathrm{H}_E(M_{\mathrm{in}})$ factors through $\mathrm{Ext}_A^1(E, \tau_A Y) = 0$, by Lemma 3.14(2). The injection into $\mathrm{H}_E(\theta_B Y)$ therefore provides a summand W of $i^! M_{\mathrm{out}}$ with $\mathrm{H}_E(W) \neq 0$. Assign to the sequence one such pair (W, Y) , for which $c_{WY} > 0$.

Case 2: $\mathrm{G}_E(Y) \neq 0$. The dual argument applies to the sequence starting at X . Write $\theta_A^- X = M'_{\mathrm{in}} \oplus M'_{\mathrm{out}}$ as above. Since ξ is also the almost split sequence of $\mathbf{mod} B$ starting at X , the dual part of Proposition 3.8 gives $\theta_B Y = M'_{\mathrm{in}} \oplus i^* M'_{\mathrm{out}}$ and $[i^* M'_{\mathrm{out}} : W]_{\oplus} = c_{XW}$. The map $\mathrm{G}_E(Y) \rightarrow \mathrm{G}_E(M'_{\mathrm{in}})$ factors through $\mathrm{Ext}_A^1(\tau_A^- X, E)$, which vanishes by the second isomorphism in (2.2) because $\mathrm{Hom}_A(E, X) = 0$. This assigns a pair (X, W) with $c_{XW} > 0$ and $\mathrm{G}_E(W) \neq 0$.

The pairs of Case 1 have distinct targets Y , and those of Case 2 distinct sources X , by uniqueness of almost split sequences. A pair (V', W') chosen in both cases would have $\mathrm{H}_E(V') \neq 0$ and $\mathrm{G}_E(W') \neq 0$, contrary to (3.9). Thus all assigned pairs are distinct, and their number is at most $\sum_{V, W} c_{VW} = c$. $\square$

Corollary 3.16. *If A is representation-finite and representation-directed and e is a primitive idempotent, then $r \leq c$.*

Proof. Here E is a single simple module. If $\mathrm{Ext}_A^1(M, N) \neq 0$ for indecomposables, the second isomorphism in (2.2) gives a nonzero map $N \rightarrow \tau_A M$. Following it by the two maps through a middle summand Z of the almost split sequence ending at M gives nonzero maps $N \rightarrow \tau_A M \rightarrow Z \rightarrow M$ between indecomposables. Removing an isomorphism step if present leaves nonzero nonisomorphisms. For $W \in \mathrm{ind} B$, nonzero $\mathrm{Ext}_A^1(E, W)$ and $\mathrm{Ext}_A^1(W, E)$ would therefore give a cycle. Condition (3.8) holds, and Theorem 3.15 applies. $\square$

In Example 3.13, the assignment of Theorem 3.15 can be computed directly.

Example 3.17. Keep $A = (k(1 \rightarrow 2 \leftarrow 3))^{\text{op}}$ and $e = e_3$, so that $E = S_3$. Since S_3 is injective, $G_E = 0$, and (3.8) holds. Since $\text{rad } P_3 = S_2$, we have $H_E(S_2) = \text{Ext}_A^1(S_3, S_2) \neq 0$. Thus the almost split sequence $0 \rightarrow S_2 \rightarrow P_1 \rightarrow S_1 \rightarrow 0$ of $\text{mod } B$ has $H_E(X) \neq 0$ for $X = S_2$. In the proof of Theorem 3.15, $\theta_A S_1 = I_2$ has no summand in $\text{mod } B$, and $i^! I_2 = P_1$. The proof therefore assigns to this sequence the pair (P_1, S_1) , with $c_{P_1 S_1} = 1$.

4. POSET REPRESENTATIONS AND DELETION OF A SIMPLE MODULE

We compute the magnitude of a representation-finite category of poset representations. The equivalence of Ringel and Vossieck (Proposition 4.15) then applies this result to the ideal quotient associated with a primitive idempotent in a directed algebra. Combined with Theorem 3.12 and Corollary 3.16, it proves Proposition 4.19, which compares $\sigma(A)$ with $\sigma(A/AeA)$.

Three results on the quotient use the work of Ringel and Vossieck. Proposition 4.12, on $\dim_k Xe$ at the τ -projectives and τ -injectives X of $\mathcal{D}$, follows from their bound on $\dim_k Xe$ for indecomposable A -modules. They proved the equivalence $\mathcal{D} \simeq T\text{-Sp}$ (Proposition 4.15). The inequality $\chi(\mathcal{D}) \geq 1$ (Corollary 4.17) follows from this equivalence and Theorem 4.5, whose proof uses their results on T -spaces. Appendix A proves all three directly, so Proposition 4.19 does not depend on [21].

4.1. Magnitude of poset representations. We first define representations of a poset by subspaces and compute the magnitude of the category they form.

Definition 4.1. For a finite poset T , a *representation of T by subspaces*, or T -space, is a finite-dimensional vector space W , its *total space*, with subspaces W_t , its *indexed subspaces*, such that $W_s \subseteq W_t$ whenever $s \leq t$. A morphism $(W; W_t) \rightarrow (W'; W'_t)$ is a linear map $f : W \rightarrow W'$ such that $f(W_t) \subseteq W'_t$ for every $t \in T$. Write $T\text{-Sp}$ for this category. A subset $U \subseteq T$ is *upward-closed* if $s \in U$ and $s \leq t$ imply $t \in U$. For such U , let K_U have total space k , with subspace $(K_U)_t = k$ for $t \in U$ and $(K_U)_t = 0$ otherwise.

In the proof of Theorem 4.5 below, we construct a function h on the indecomposable T -spaces that increases by one along arrows and satisfies $h(X) = h(\tau^+ X) + 2$ for nonzero translates. The following lemma computes the number of arrows, and hence the magnitude, from such a function.

Lemma 4.2. *Let $\mathcal{C}$ be a Hom-finite strict τ -category with finitely many indecomposable isomorphism classes. Suppose there are indecomposables X_0, X_1 and a function $h : \text{ind } \mathcal{C} \rightarrow \mathbb{Z}$ with the following properties.*

- (1) *The object X_0 is τ -projective and has no incoming arrow, and every other τ -projective has exactly one incoming arrow, counted with multiplicity. Dually, X_1 is τ -injective and has no outgoing arrow, and every other τ -injective has exactly one outgoing arrow.*
- (2) *One has $h(X_0) = 0$, $h(Y) = h(X) + 1$ for every arrow $X \rightarrow Y$, and $h(X) = h(\tau_C^+ X) + 2$ whenever $\tau_C^+ X \neq 0$.*

Then, with $L = h(X_1)$,

$$a(\mathcal{C}) = 2|\text{ind } \mathcal{C}| - L - 2, \quad \chi(\mathcal{C}) = 1 + L - (|\text{ind}_1^+ \mathcal{C}| - 1).$$

Proof. By Proposition 2.7(3), τ_C^+ and τ_C^- are inverse bijections between $\text{ind } \mathcal{C} \setminus \text{ind}_1^+ \mathcal{C}$ and $\text{ind } \mathcal{C} \setminus \text{ind}_1^- \mathcal{C}$, so $|\text{ind}_1^- \mathcal{C}| = |\text{ind}_1^+ \mathcal{C}|$. For $X \notin \text{ind}_1^+ \mathcal{C}$, the right τ -sequence at X is the left τ -sequence at $\tau_C^+ X$, by part (2) of the same proposition. Lemma 2.9 therefore gives

$$\text{indeg}_{\mathcal{C}}(X) = \text{outdeg}_{\mathcal{C}}(\tau_C^+ X). \quad (4.1)$$

Let h_+ and h_- be the sums of h over $\text{ind}_1^+ \mathcal{C}$ and $\text{ind}_1^- \mathcal{C}$. Summing $h(X) - h(\tau_C^+ X) = 2$ over $X \notin \text{ind}_1^+ \mathcal{C}$ gives $h_- - h_+ = 2(|\text{ind } \mathcal{C}| - |\text{ind}_1^+ \mathcal{C}|)$.

Each arrow raises h by one, so $a(\mathcal{C}) = \sum_X h(X)(\text{indeg}_{\mathcal{C}}(X) - \text{outdeg}_{\mathcal{C}}(X))$. In the incoming sum, the τ -projectives contribute h_+ , by (1) and $h(X_0) = 0$. For the other X , write $h(X) = h(\tau_C^+ X) + 2$, use (4.1), and reindex by $Y = \tau_C^+ X$, which runs over $\text{ind } \mathcal{C} \setminus \text{ind}_1^- \mathcal{C}$. In the outgoing sum, the

τ -injectives contribute $h_- - L$, by (1). Also by (1), the τ -injectives other than X_1 have one outgoing arrow each, and $|\text{ind}_1^- \mathcal{C}| = |\text{ind}_1^+ \mathcal{C}|$, so $\sum_{Y \notin \text{ind}_1^- \mathcal{C}} \text{outdeg}_{\mathcal{C}}(Y) = a(\mathcal{C}) - (|\text{ind}_1^+ \mathcal{C}| - 1)$. Hence

$$\begin{aligned} a(\mathcal{C}) &= h_+ + \sum_{Y \notin \text{ind}_1^- \mathcal{C}} (h(Y) + 2) \text{outdeg}_{\mathcal{C}}(Y) - \sum_{Y \notin \text{ind}_1^- \mathcal{C}} h(Y) \text{outdeg}_{\mathcal{C}}(Y) - (h_- - L) \\ &= 2(a(\mathcal{C}) - (|\text{ind}_1^+ \mathcal{C}| - 1)) + h_+ - h_- + L \\ &= 2(a(\mathcal{C}) - |\text{ind}_1^+ \mathcal{C}| + 1) - 2(|\text{ind} \mathcal{C}| - |\text{ind}_1^+ \mathcal{C}|) + L, \end{aligned}$$

and solving gives $a(\mathcal{C}) = 2|\text{ind} \mathcal{C}| - L - 2$. Proposition 2.14 gives $\chi(\mathcal{C}) = 2|\text{ind} \mathcal{C}| - a(\mathcal{C}) - |\text{ind}_1^+ \mathcal{C}| = 1 + L - (|\text{ind}_1^+ \mathcal{C}| - 1)$. $\square$

We recall translation quivers, used in the next lemma, and their mesh categories, used in Subsection 5.3.

Definition 4.3. A *translation quiver* is a locally finite quiver without loops together with an injective map τ , the translation, from a subset of the vertex set to the vertex set, such that for every vertex X in its domain and every vertex Y , the number of arrows $Y \rightarrow X$ equals the number of arrows $\tau X \rightarrow Y$. The k -linear path category of a translation quiver has its vertices as objects; its morphisms are linear combinations of oriented paths, including the stationary identity paths, and composition is concatenation. A *mesh presentation* is a choice, at each vertex X with a translate and each vertex Y , of a bijection between the arrows $\tau X \rightarrow Y$ and the arrows $Y \rightarrow X$; we write the paired arrows as $b_{Y,j} : \tau X \rightarrow Y$ and $a_{Y,j} : Y \rightarrow X$. The *mesh relation* at X is

$$\sum_{Y,j} a_{Y,j} b_{Y,j} = 0.$$

The *mesh category* is the path category modulo these relations.

A function h as in Lemma 4.2 is obtained from path lengths by the following lemma of Ringel and Vossieck.

Lemma 4.4 ([21, Section 7, Lemma, pp. 234–235]). *Let Γ be a translation quiver with a unique source, in which only finitely many vertices have no translate. Suppose that each vertex without translate has at most one immediate predecessor, that is, a vertex with an arrow to it. Then all paths in Γ from the source to a fixed vertex have the same length.*

Theorem 4.5. *Let T be a finite poset such that $T\text{-Sp}$ is representation-finite. Then its Hom matrix is invertible. Every path of irreducible maps from $K_{\emptyset}$ to K_T has the same length L , and*

$$\chi(T\text{-Sp}) = 1 + L - |T| \geq 1. \quad (4.2)$$

The following conditions are equivalent:

- (1) $\chi(T\text{-Sp}) = 1$;
- (2) every indecomposable T -space has one-dimensional total space;
- (3) the width of T , the largest size of an antichain, is at most two.

Proof. We use the following results of Ringel and Vossieck on a finite representation-finite poset T .

- (a) The category $T\text{-Sp}$ has almost split sequences, that is, sequences $X \rightarrow Y \rightarrow Z$ of T -spaces whose first map is minimal left almost split and whose last map is minimal right almost split. They are exact on total spaces and on all indexed subspaces [21, Section 1, p. 219, and Section 7, p. 234]. Its Auslander–Reiten quiver is finite, since $T\text{-Sp}$ is representation-finite.
- (b) The object $K_{\emptyset}$ is the unique source and K_T the unique sink of the Auslander–Reiten quiver. The vertices without translate are $K_{\emptyset}$ and $K_{\{s:t \leq s\}}$ for $t \in T$, and the minimal right almost split maps ending at them are monomorphisms. Each of them other than $K_{\emptyset}$ is the target of exactly one arrow. Dually, K_T has no inverse translate, and every other vertex without inverse translate is the source of exactly one arrow [21, Section 7, pp. 234–235].
- (c) The common length L of the paths from $K_{\emptyset}$ to K_T satisfies $L \geq |T|$ [21, Section 8, p. 237]. Moreover, $L = |T|$ if and only if every indecomposable has one-dimensional total space, if and only if T has width at most two [21, Section 8, Proposition, p. 237].

We now verify that $T\text{-Sp}$ satisfies the hypotheses of Lemma 4.2.

A sequence that is exact on total spaces and on all indexed subspaces is a kernel–cokernel pair in $T\text{-Sp}$: a map into its middle term killed by the last map factors through the first map on total spaces, and exactness on each indexed subspace shows that the factorization is a morphism of T -spaces. Hence each almost split sequence is a right and a left τ -sequence with monic first map and epic last map. At a vertex without translate, the monic minimal right almost split map of (b) gives a right τ -sequence with zero first term.

For the dual statement, note that $(W; W_t) \mapsto (W^*; W_t^\perp)$ is a contravariant equivalence from T -spaces to T^{op} -spaces. Since $\text{Hom}(K_\emptyset, W)$ is the total space of W and $\text{Hom}(W, K_T)$ is its dual, the monomorphisms and epimorphisms of T -spaces are exactly the maps that are injective, respectively surjective, on total spaces. The duality exchanges these two classes. Applied to T^{op} , (b) therefore shows that the minimal left almost split map starting at a vertex without inverse translate is epic. It gives a left τ -sequence with zero last term. Thus $T\text{-Sp}$ is a strict τ -category, its τ -projectives are the vertices listed in (b), and $|\text{ind}_1^+ T\text{-Sp}| = |T| + 1$. By Proposition 2.7(2) and Lemma 2.9, the number of arrows $Y \rightarrow X$ equals the number of arrows $\tau_{T\text{-Sp}}^+ X \rightarrow Y$ whenever $\tau_{T\text{-Sp}}^+ X \neq 0$. Hence the irreducible quiver of $T\text{-Sp}$ with the translation $\tau_{T\text{-Sp}}^+$, which is its Auslander–Reiten quiver, is a translation quiver.

Every indecomposable $X \not\cong K_\emptyset$ is the target of a path from $K_\emptyset$: the space $\text{Hom}(K_\emptyset, X)$ is the total space of X , which is nonzero, and a nonzero map $K_\emptyset \rightarrow X$ is radical, hence a linear combination of paths of irreducible maps by Lemma 2.10. By (b) and Lemma 4.4, the length of the paths from $K_\emptyset$ therefore defines a function h with $h(K_\emptyset) = 0$ that increases by one along each arrow. If $\tau_{T\text{-Sp}}^+ X \neq 0$, strictness gives a nonzero middle term, so X and $\tau_{T\text{-Sp}}^+ X$ are joined by paths of length two and $h(X) = h(\tau_{T\text{-Sp}}^+ X) + 2$. Lemma 4.2 now applies with $X_0 = K_\emptyset$, $X_1 = K_T$, and $h(K_T) = L$. It gives $a(T\text{-Sp}) = 2|\text{ind } T\text{-Sp}| - L - 2$ and $\chi(T\text{-Sp}) = 1 + L - |T|$. Now (c) gives $\chi(T\text{-Sp}) \geq 1$.

(1) $\Leftrightarrow$ (2): By (4.2), $\chi(T\text{-Sp}) = 1$ if and only if $L = |T|$, and by (c) this holds if and only if every indecomposable has one-dimensional total space.

(2) $\Leftrightarrow$ (3): This is part of (c). $\square$

4.2. Ideal quotients of representation-directed algebras. We now return to modules. The rest of this section compares an algebra with its quotient by a primitive idempotent.

Definition 4.6. For a primitive idempotent e of an algebra A , the algebra A/AeA is called the *primitive deletion* of e .

For a representation-directed algebra and a primitive idempotent, we fix the following notation for the associated ideal quotient.

Setting 4.7. Throughout Subsections 4.2–4.6 and Appendix A, except in statements and examples that name another algebra, let A be basic, representation-finite, and representation-directed. Fix a primitive idempotent e , and put

$$B = A/AeA, \quad \mathcal{D} = (\text{mod } A)/[\text{mod } B], \quad P = eA, \quad E = P/\text{rad } P, \quad I = D_k(Ae).$$

Then P is the projective cover of the simple module E , I is its injective envelope, and $\mathcal{D}$ is the ideal quotient of Setting 3.3 for this idempotent. Thus $E, P = P_E$, and $I = I_E$ agree with the notation of Subsection 3.5. Put $d_X = \dim_k Xe = [X : E]$, where $[X : E]$ is composition multiplicity. The equality $\dim_k Xe = [X : E]$ follows by applying the exact functor $(-)e$ to a composition series.

The following lemma shows that B is representation-finite, as assumed from Subsection 3.4 on, and representation-directed, as needed for induction on primitive deletions.

Lemma 4.8. *The algebra B is representation-finite and representation-directed.*

Proof. The inclusion of $\text{mod } B$ in $\text{mod } A$ is full (Setting 3.3), so it preserves indecomposables and their Hom spaces. Hence B is representation-finite and directed. $\square$

4.3. Source, sink, and translation of the quotient. Lemma 3.4 and Proposition 3.2 already describe the maps and the indecomposables of $\mathcal{D}$. For a directed algebra, we also show that the indecomposables of $\mathcal{D}$ have endomorphism ring k , that $\mathcal{D}$ is directed, and that P and I are the unique source and sink of its irreducible quiver. We identify the indecomposables of $\mathcal{D}$ with the chosen A -module representatives with $Xe \neq 0$, writing X for πX when it occurs as an object of $\mathcal{D}$.

Lemma 4.9. *Evaluation at e and the quotient functor π have the following properties.*

- (1) *Every indecomposable A -module has endomorphism ring k . The category $\mathcal{D}$ is Krull–Schmidt and directed. Its indecomposables are the images of the indecomposable A -modules with $d_X > 0$; they remain distinct and have endomorphism ring k .*
- (2) *For every object X ,*

$$\mathcal{D}(P, X) = \text{Hom}_A(P, X) \cong Xe, \quad \mathcal{D}(X, I) = \text{Hom}_A(X, I) \cong D_k(Xe). \quad (4.3)$$

The object P is the unique source and I the unique sink of the irreducible quiver of $\mathcal{D}$.

Proof. (1) Directedness gives $\text{End}_A(X) = k$ for every indecomposable: a nonzero noninvertible endomorphism would be a cycle. Proposition 3.2 identifies the indecomposables of $\mathcal{D}$ and their local endomorphism rings, which are therefore k . A cycle of nonzero nonisomorphisms in $\mathcal{D}$ would lift to one over A , so $\mathcal{D}$ is directed.

(2) The identifications in (4.3) hold because $\text{Hom}_A(P, \text{mod } B) = 0$ and $\text{Hom}_A(\text{mod } B, I) = 0$. Every indecomposable of $\mathcal{D}$ therefore receives a nonzero map from P and maps nontrivially to I . An incoming arrow at P , or an outgoing arrow at I , would give a cycle. Expanding the nonzero maps from P and to I into paths by Lemma 2.10 gives an incoming arrow at every $X \neq P$ and an outgoing arrow at every $X \neq I$, by taking the last and first arrows of these paths, respectively. $\square$

Part (2) is also observed in [21, Section 10].

By Proposition 3.7, $\mathcal{D}$ is a strict τ -category whose right and left τ -sequences at X , in the notation introduced before Proposition 2.7, are the images of the right and left τ -sequences of $\text{mod } A$ ending and starting at X (Example 2.8).

Definition 4.10. Write $\Gamma_{\mathcal{D}}$ for the irreducible quiver of $\mathcal{D}$ equipped with the translation $\tau_{\mathcal{D}}^+$, displayed by dotted arrows $X \dashrightarrow \tau_{\mathcal{D}}^+ X$.

Since $\mathcal{D}$ is directed (Lemma 4.9(1)), $\Gamma_{\mathcal{D}}$ has no loops, and $\tau_{\mathcal{D}}^+$ is injective by Proposition 2.7(3). By Proposition 2.7(2) and Lemma 2.9, the number of arrows $Y \rightarrow X$ equals the number of arrows $\tau_{\mathcal{D}}^+ X \rightarrow Y$ whenever $\tau_{\mathcal{D}}^+ X \neq 0$. Thus $\Gamma_{\mathcal{D}}$ is a translation quiver (Definition 4.3).

Lemma 4.11. *Let X be an indecomposable A -module with $Xe \neq 0$. Its image is τ -projective in $\mathcal{D}$ precisely when X is projective over A or $\tau_A X$ belongs to $\text{mod } B$. Dually, it is τ -injective precisely when X is injective over A or $\tau_A^- X$ belongs to $\text{mod } B$.*

Proof. By Proposition 3.7, $\tau_{\mathcal{D}}^+(\pi X) \cong \pi(\tau_A X)$, and π kills exactly the modules in $\text{mod } B$ (Proposition 3.2). The dual statement follows in the same way. $\square$

4.4. Composition multiplicities at τ -projectives and τ -injectives. We next bound the multiplicity d_X of E at the τ -projectives and τ -injectives of $\mathcal{D}$. The bound shows that an almost split sequence with exactly one end term in $\text{mod } B$ has exactly one middle summand outside it, which removes the term δ in Theorem 3.12.

Proposition 4.12 ([21, Section 10, Corollary, p. 243]). *In Setting 4.7, the following hold.*

- (1) *Every τ -projective and τ -injective of $\mathcal{D}$ has $d_X = 1$.*
- (2) *For a nonprojective indecomposable A -module Y ,*

$$(\tau_A Y)e = 0 \implies d_Y \leq 1, \quad Ye = 0 \implies d_{\tau_A Y} \leq 1. \quad (4.4)$$

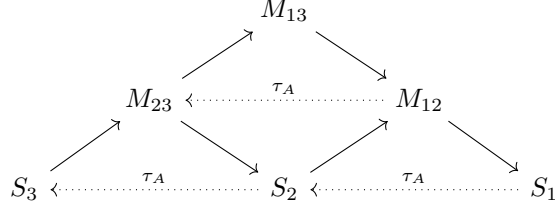
Proof. Ringel and Vossieck show that over a representation-directed algebra, every indecomposable projective or injective module Z has $d_Z \leq 1$, and $|d_Z - d_{\tau_A Z}| \leq 1$ for every nonprojective indecomposable Z [21, Section 10, Corollary, p. 243]; they state that it is due to v. Höhne. Part (2) is the case $Z = Y$ of this bound. For part (1), a τ -projective X of $\mathcal{D}$ is projective over A or satisfies

$(\tau_A X)e = 0$, by Lemma 4.11. In both cases $d_X \leq 1$, and $d_X \geq 1$ since $X \notin \text{mod } B$. The τ -injectives are dual. $\square$

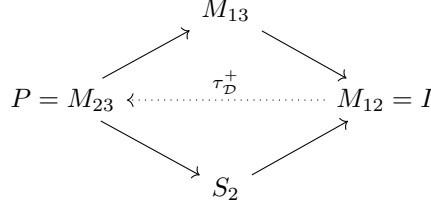
Appendix A.1 gives a proof that does not use [21].

In the following example the multiplicities of Proposition 4.12 can be computed directly; it is used again for the poset construction in Example 4.16.

Example 4.13. Let Q be $1 \rightarrow 2 \rightarrow 3$ and $A = (kQ)^{\text{op}}$. Write M_{ij} for the interval representation on vertices $i, \dots, j$, with identity maps between nonzero coordinates, and $S_i = M_{ii}$. The Auslander–Reiten quiver of A is



Take e to be the vertex idempotent at 2, so $E = S_2$. Then $B \cong k \times k$, whose indecomposables S_1, S_3 become zero in $\mathcal{D}$. The quotient has translation quiver



The only nonzero translate is $\tau_{\mathcal{D}}^+ M_{12} = M_{23}$. In particular, the right τ -sequence at S_2 has first term zero because its first term over A is S_3 . Hence $\text{ind}_1^+ \mathcal{D} = \{M_{23}, M_{13}, S_2\}$. Each has dimension one at vertex 2, as does $I = M_{12}$.

4.5. Identification with poset representations. We now construct the poset T and the functor identifying $\mathcal{D}$ with the category of T -spaces. The functor G below is the functor of Ringel and Vossieck [21, Section 1, Theorem 3, p. 221, and Section 7], written in terms of subspaces; Appendix A.3 proves the equivalence by a different route.

Construction 4.14. Index the τ -projectives other than P by a set T , choosing indecomposable A -module representatives Q_t . Define

$$s \leq t \iff \mathcal{D}(Q_t, Q_s) \neq 0.$$

This order is chosen so that Q_t corresponds to $K_{\{s:t \leq s\}}$ (Proposition 4.15). Let $F(X) = Xe$ be the evaluation functor of Lemma 3.4. Since $eAe \cong \text{End}_A(P) = k$ by Lemma 4.9(1),(2), identify $F(P) = eAe$ with k , using e as basis. By Proposition 4.12(1), each $F(Q_t)$ is one-dimensional. Choose a basis vector of each $F(Q_t)$, and let $u_t \in \mathcal{D}(P, Q_t)$ be the map corresponding to it under (4.3), so that its evaluation sends 1 to that basis vector. Identify $\mathcal{D}(P, X)$ with $F(X)$ by (4.3), and set

$$F_t(X) = \text{Im}(\mathcal{D}(Q_t, X) \longrightarrow \mathcal{D}(P, X), f \longmapsto fu_t), \quad G(X) = (F(X); F_t(X)). \quad (4.5)$$

On morphisms, G is evaluation at e . The subspace $F_t(X)$ does not change if u_t is rescaled, and G on morphisms does not depend on the chosen bases. The bases are used only in Appendix A.3, to realize an arbitrary T -space by a module.

Proposition 4.15 ([21, Section 10, Theorem, p. 243]). *Construction 4.14 defines a finite poset and a k -linear equivalence $G : \mathcal{D} \xrightarrow{\sim} T\text{-Sp}$. Moreover,*

$$G(P) = K_{\emptyset}, \quad G(Q_t) = K_{\{s:t \leq s\}}, \quad G(I) = K_T, \quad |T| = |\text{ind}_1^+ \mathcal{D}| - 1.$$

We compute the equivalence on the four objects of Example 4.13.

Example 4.16. For Example 4.13, let $Q_{t_1} = M_{13}$ and $Q_{t_2} = S_2$. They have no nonzero maps between them in $\mathcal{D}$, so T is a two-element antichain. Evaluation and the two indexed subspaces give

X	$F(X)$	$F_{t_1}(X)$	$F_{t_2}(X)$	$G(X)$
M_{23}	k	0	0	$K_\emptyset$
M_{13}	k	k	0	$K_{\{t_1\}}$
S_2	k	0	k	$K_{\{t_2\}}$
M_{12}	k	k	k	K_T

Each arrow of the translation quiver of $\mathcal{D}$ in Example 4.13 is the identity on total spaces. The paths of irreducible maps from $K_\emptyset$ to K_T have length $L = 2 = |T|$, so $\chi(\mathcal{D}) = 1$ by Theorem 4.5.

Corollary 4.17. *One has $\chi(\mathcal{D}) \geq 1$, and the following are equivalent.*

- (1) $\chi(\mathcal{D}) = 1$.
- (2) $d_X = 1$ for every $X \in \text{ind } \mathcal{D}$.
- (3) T has width at most two.

Proof. The equivalence of Proposition 4.15 sends X to a T -space with total space Xe , and $T\text{-Sp}$ is representation-finite because $\mathcal{D}$ is. Thus $\chi(\mathcal{D}) = \chi(T\text{-Sp})$, and conditions (1)–(3) are conditions (1)–(3) of Theorem 4.5, which proves the corollary. $\square$

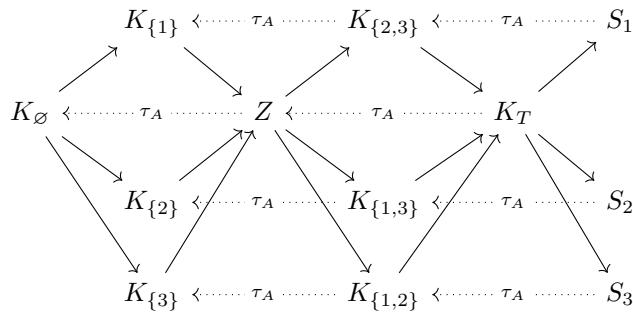
Appendix A.4 proves this corollary directly, without Theorem 4.5.

In the following D_4 example, $\chi(\mathcal{D}) > 1$, in contrast with Example 4.16.

Example 4.18. Let Q be the Dynkin quiver of type D_4 with arrows $1 \rightarrow 0$, $2 \rightarrow 0$, $3 \rightarrow 0$, and put $A = (kQ)^{\text{op}}$. Take e at vertex 0. The three τ -projectives other than $P = S_0$ are the projectives supported on $\{0, i\}$, so T is a three-element antichain. We identify the element of T indexing the projective supported on $\{0, i\}$ with i , so that $T = \{1, 2, 3\}$.

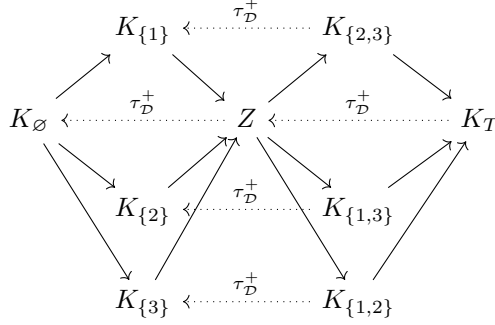
Its representations are triples of subspaces of a vector space. Besides the eight one-dimensional representations K_U , for $U \subseteq T$, there is an indecomposable Z with total space k^2 and indexed lines $k(1, 0)$, $k(0, 1)$, and $k(1, 1)$. The A -module K_U has space k at 0 and at the vertices in U , with identity maps between nonzero spaces. The module Z has space k^2 at 0, and its three arrow maps are the inclusions of the lines $k(1, 0)$, $k(0, 1)$, and $k(1, 1)$. The functor G of Proposition 4.15 sends these modules to the T -spaces with the same names.

Together with the simples S_1, S_2, S_3 , these are the indecomposable A -modules, and knitting gives the Auslander–Reiten quiver of A :



The projectives are $K_\emptyset, K_{\{1\}}, K_{\{2\}}, K_{\{3\}}$. The arrow $K_{\{i\}} \rightarrow Z$ is the inclusion of the i -th indexed line, the arrow $Z \rightarrow K_{T \setminus \{i\}}$ is the quotient map with kernel that line, and the other arrows between the modules K_U are identities on total spaces. The almost split sequences ending at Z , $K_{\{i,j\}}$, K_T , and S_i have middle terms $\bigoplus_i K_{\{i\}}$, Z , $\bigoplus_{|U|=2} K_U$, and K_T , respectively.

The modules S_1, S_2, S_3 become zero in $\mathcal{D}$. By Propositions 3.2 and 3.7, the remaining arrows and the images of the translations give the translation quiver $\Gamma_{\mathcal{D}}$:



The quotient has nine indecomposables, twelve arrows, and four τ -projectives. Thus (2.4) gives

$$\chi(\mathcal{D}) = 2 \cdot 9 - 12 - 4 = 2, \quad L = 4 > |T| = 3.$$

4.6. Magnitude under deletion of a simple module. We now apply Theorem 3.12 and Corollary 3.16 to the Serre subcategory $\text{mod } B$.

Proposition 4.19. *In Setting 4.7, with c, r as in Definition 3.10, one has*

$$\sigma(A) - \sigma(B) = \chi(\mathcal{D}) - 1 + c - r \geq 0. \quad (4.6)$$

If equality holds in $\sigma(A) \geq \sigma(B)$, then $\dim_k X e \leq 1$ for every indecomposable A -module X .

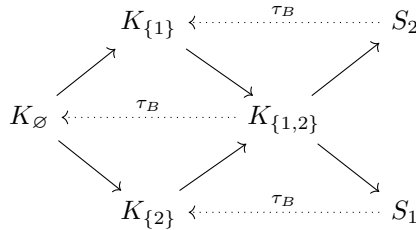
Proof. By Lemma 4.8, $\text{mod } B$ is a representation-finite Serre subcategory, so Section 3 applies. In an almost split sequence with exactly one end term in $\text{mod } B$, Proposition 4.12(2) gives $d_X \leq 1$ for the other end term X , and $d_X \geq 1$ since $X \notin \text{mod } B$. The sequence stays exact after applying $(-)e$, and its end term in $\text{mod } B$ contributes zero. So $d_X = 1$ is the sum of $d_Z \geq 1$ over the middle summands Z outside $\text{mod } B$, counted with multiplicity. Hence there is exactly one middle summand outside $\text{mod } B$, with multiplicity one. Thus $\delta = 0$ in Theorem 3.12. Since $|A| = |B| + 1$, its identity becomes

$$\sigma(A) - \sigma(B) = \chi(\mathcal{D}) - 1 + c - r.$$

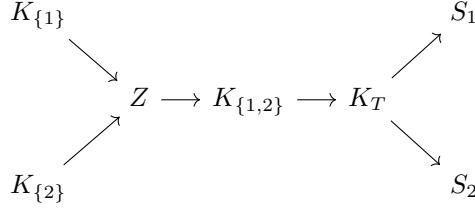
Corollaries 4.17 and 3.16 make the two terms nonnegative. Equality implies $\chi(\mathcal{D}) = 1$, so $d_X = 1$ for every $X \in \text{ind } \mathcal{D}$ by Corollary 4.17(1) $\Rightarrow$ (2); the other indecomposable A -modules have $d_X = 0$. $\square$

Deleting a leaf in Example 4.18 gives a strict inequality $r < c$. We compute c and r explicitly.

Example 4.20. Keep $A = (kQ)^{\text{op}}$ with arrows $i \rightarrow 0$, $i = 1, 2, 3$, from Example 4.18. Write e_i for the idempotent at vertex i , and now take $E = S_3$ and $B = A/Ae_3A$. Keep the notation K_U, Z from that example. The indecomposable B -modules are $K_{\emptyset}, K_{\{1\}}, K_{\{2\}}, K_{\{1,2\}}, S_1, S_2$. The Auslander–Reiten quiver Γ_B is



The two arrows out of $K_\emptyset$ are irreducible over A . The other four are not. Their factorizations in $\text{mod } A$ are shown in the following diagram of maps:



Each map in this diagram is irreducible in $\text{mod } A$. The two composites through Z and the two through K_T give the other four arrows of Γ_B : both intermediate modules have nonzero space at vertex 3, so they are not B -modules. Each of these four arrows contributes one to c , and there are no other arrows in Γ_B . Thus $c = 4$.

The three almost split sequences of $\text{mod } B$ are

$$\begin{aligned}
 0 \rightarrow K_\emptyset \rightarrow K_{\{1\}} \oplus K_{\{2\}} \rightarrow K_{\{1,2\}} \rightarrow 0, \\
 0 \rightarrow K_{\{2\}} \rightarrow K_{\{1,2\}} \rightarrow S_1 \rightarrow 0, \quad 0 \rightarrow K_{\{1\}} \rightarrow K_{\{1,2\}} \rightarrow S_2 \rightarrow 0.
 \end{aligned}$$

None is almost split over A : the almost split sequences over A with the same last terms start at $K_{\{3\}}, K_{\{2,3\}}, K_{\{1,3\}}$, respectively. Hence $r = 3$, and $c - r = 1$.

For the functors of Subsection 3.5, the projective resolution $0 \rightarrow S_0 \rightarrow K_{\{3\}} \rightarrow S_3 \rightarrow 0$ gives $H_E(X) = \text{Ext}_A^1(S_3, X) \cong X e_0$ for $X \in \text{mod } B$. The simple S_3 is injective, so $G_E = 0$. Theorem 3.15 therefore assigns to each of the three sequences a pair (W, Y) with $c_{WY} > 0$, where Y is its last term. The sequence ending at $K_{\{1,2\}}$ has two possible choices, from $K_{\{1\}}$ and $K_{\{2\}}$; the other two sequences have one each. This displays the strict inequality $r = 3 < 4 = c$. Example 4.21 combines these counts with those of $\mathcal{D}$ to check (4.6).

The two deletions in our D_4 example distinguish the two nonnegative terms in (4.6).

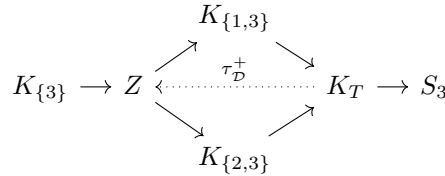
Example 4.21. The Auslander–Reiten quiver of $\text{mod } A$, displayed in Example 4.18, has twelve indecomposables and fifteen arrows. Thus, by Lemma 2.16,

$$|\text{ind } A| = 12, \quad |A| = 4, \quad a(A) = 15, \quad \chi(\text{mod } A) = 5, \quad \sigma(A) = 1.$$

Deleting vertex 0 gives $B_0 \cong k^3$. Deleting vertex 3 gives the path algebra B_3 of Example 4.20. The complete counts are

e	$ \text{ind } B $	$ B $	$a(B)$	$\chi(\mathcal{D}) - 1$	c	r	$\sigma(A) - \sigma(B)$
e_0	3	3	0	1	0	0	1
e_3	6	3	6	0	4	3	1

For the second row, the associated translation quiver $\Gamma_{\mathcal{D}}$ is



Thus it has six indecomposables and six solid arrows. Only K_T has a nonzero $\tau_{\mathcal{D}}^+$ -translate, namely Z . Hence there are five τ -projectives, and (2.4) gives $\chi(\mathcal{D}) = 2 \cdot 6 - 6 - 5 = 1$. The strict increase comes entirely from $\chi(\mathcal{D}) - 1$ in the first row and from $c - r$ in the second.

4.7. Magnitude of representation-directed algebras. Applying Proposition 4.19 to one simple module at a time bounds the surplus of a representation-directed algebra from below, and zero surplus leads to biserial algebras. We recall the classes of algebras that enter the proof of Theorem A(2).

Definition 4.22. A basic algebra is *special biserial* if it has a presentation $kQ/\mathfrak{r}$, with $R_Q^m \subseteq \mathfrak{r} \subseteq R_Q^2$ for the arrow ideal R_Q and some $m \geq 2$, satisfying two conditions: at most two arrows enter and at most two leave each vertex; and for each arrow α , there is at most one arrow β for which $\beta\alpha$ is defined and does not belong to $\mathfrak{r}$, and at most one arrow γ with the same property for $\alpha\gamma$. These conditions are symmetric under replacing Q by its opposite quiver.

Definition 4.23. A module is *uniserial* if its submodules are totally ordered by inclusion. An algebra is *biserial* if the radical of every indecomposable projective, on either side, is a sum of two uniserial submodules whose intersection is simple or zero. A module over a basic algebra is *thin* if each simple occurs in it with multiplicity at most one, equivalently if $\dim_k Xe \leq 1$ at every primitive idempotent in a complete orthogonal set.

If a representation-directed algebra has zero surplus, then all its indecomposable modules are thin (Corollary 4.26 below). Two results show that such an algebra is special biserial. The first is due to Pogorzały and Skowroński.

Proposition 4.24 ([20, Proposition 1]). *If every indecomposable module over a basic finite-dimensional algebra is thin, then the algebra is representation-finite and biserial.*

The second is due to Skowroński and Waschbüsch.

Theorem 4.25 ([22, p. 173 and Lemma 2]; see also [20, Theorem 2]). *Every basic representation-finite biserial algebra is special biserial.*

Corollary 4.26. *For a representation-finite representation-directed algebra A , $\sigma(A) \geq 0$. If A is basic and $\sigma(A) = 0$, then every indecomposable A -module is thin, and A is biserial and hence special biserial.*

Proof. By Lemma 2.18 we may assume that A is basic. The algebras obtained by successive primitive deletions remain basic, and by Lemma 4.8 they remain representation-finite and representation-directed. They end at the zero algebra, whose surplus is zero. Proposition 4.19 therefore proves nonnegativity. If $\sigma(A) = 0$, delete any primitive idempotent e first. Since $\sigma(B) \geq 0$ for $B = A/AeA$, we have $\sigma(A) = \sigma(B) = 0$, so $\dim_k Xe \leq 1$ for every indecomposable X . As e was arbitrary, all indecomposables are thin. Proposition 4.24 shows that A is biserial, and Theorem 4.25 that it is special biserial. $\square$

5. A LOWER BOUND FOR MAGNITUDE VIA GRADED INTERVAL ALGEBRAS

Corollary 4.26 proves $\chi(\text{mod } A) \geq |A|$ only for representation-directed algebras. To remove this hypothesis, we compare a graded algebra with its interval algebras, whose modules are graded modules supported in finite degree intervals. We first show that the numbers of indecomposables and of arrows of the interval algebras grow linearly with the length of the interval, with slopes the numbers of indecomposables and of arrows of the algebra. A positive grading of the Hom spaces between indecomposable modules (Subsection 5.2) then makes the interval algebras representation-directed, and standard forms have such a grading.

5.1. Graded modules supported in finite intervals. We first fix conventions for graded modules and define interval algebras.

Definition 5.1. Let $R = \bigoplus_{d \geq 0} R^{(d)}$ be a nonnegatively graded algebra. A graded right R -module is a decomposition $M = \bigoplus_{n \in \mathbb{Z}} M_n$ with $M_n R^{(d)} \subseteq M_{n+d}$. Its support consists of the integers n with $M_n \neq 0$. Write $\text{grmod } R$ for the category of finite-dimensional graded modules and maps of degree zero, and Hom_{gr} for its morphism spaces. The shift $M\langle s \rangle$ is defined by $(M\langle s \rangle)_n = M_{n-s}$. A homogeneous map of degree d sends M_n to N_{n+d} ; it has degree zero as a map $M\langle s \rangle \rightarrow N\langle t \rangle$ when $d = s - t$.

Construction 5.2. For $m \geq 0$, define the *interval algebra* R_m to be the upper triangular block algebra with blocks indexed by $0, \dots, m$,

$$(R_m)_{rs} = \begin{cases} R^{(s-r)} & r \leq s, \\ 0 & r > s. \end{cases} \quad (5.1)$$

Multiplication of blocks is induced by multiplication in R . Thus its diagonal blocks are copies of $R^{(0)}$.

Remark 5.3. Viewed as a category with objects $0, \dots, m$, the algebra R_m is the full subcategory on these objects of the smash product of R with $\mathbb{Z}$ in the sense of Cibils and Marcos [9, Definition 3.1]. The equivalence of $\mathbf{mod} R_m$ with graded modules in Proposition 5.4 below is a case of [12, Theorem 6.6].

Proposition 5.4. *The category $\mathbf{mod} R_m$ is equivalent to the full subcategory of $\mathbf{grmod} R$ supported in $[0, m]$, and $|R_m| = (m+1)|R|$. If R is representation-finite, every indecomposable R -module has a graded lift, that is, a graded module whose underlying module is isomorphic to it; this lift is unique up to shift, and R_m is representation-finite.*

Proof. The representation-finiteness of R_m is also [13, Proposition 1.1]; we include the short arguments. A right R_m -module has spaces M_r at the diagonal idempotents, with multiplication maps $M_r R^{(s-r)} \rightarrow M_s$. These give a graded R -module by setting its other components to zero. Associativity holds across the interval boundary because multiplication only increases degree: if the initial and final degrees lie in $[0, m]$, so does every intermediate degree. This construction and restriction to the interval are inverse on modules and maps. The ideal of elements of positive degree in R is nilpotent, so $|R| = |R^{(0)}|$. The strictly upper triangular ideal of R_m is also nilpotent, and its diagonal quotient is $(R^{(0)})^{m+1}$, which proves $|R_m| = (m+1)|R|$.

Gordon–Green prove that a finite-dimensional graded module is indecomposable if and only if its underlying module is indecomposable, and that two graded lifts of an indecomposable differ by shift [12, Theorems 3.2 and 4.1]. They also prove that every module over a representation-finite graded algebra is gradable [13, Theorem 4.3]. Choose one lift of each indecomposable. Only finitely many of its shifts are supported in $[0, m]$, so the equivalence proves representation-finiteness of R_m . $\square$

Theorem 5.5. *Let R be a nonnegatively graded representation-finite algebra. Choose a graded lift $\tilde{X}$ of each $X \in \mathbf{ind} R$, normalized so that the least n with $\tilde{X}_n \neq 0$ is zero, and let w_X be the greatest such n .*

(1) For $m \geq \max_X w_X$,

$$|\mathbf{ind} R_m| = \sum_{X \in \mathbf{ind} R} (m+1 - w_X). \quad (5.2)$$

(2) As $m \rightarrow \infty$,

$$a(R_m) = (m+1)a(R) + O(1), \quad (5.3)$$

$$\sigma(R_m) = (m+1)\sigma(R) + O(1). \quad (5.4)$$

Proof. (1) The shifts of $\tilde{X}$ in the interval are precisely $\tilde{X}\langle t \rangle$ with $0 \leq t \leq m - w_X$. Proposition 5.4 proves (5.2).

(2) Choose $b \geq 1$ bounding all w_X . For $X, Y \in \mathbf{ind} R$, every map $X \rightarrow Y$ is a sum of maps homogeneous with respect to the gradings of $\tilde{X}$ and $\tilde{Y}$, whose degrees lie in $[-b, b]$ because both lifts are supported in $[0, b]$.

We first compare irreducible maps in $\mathbf{mod} R$ and in $\mathbf{grmod} R$. Fix $t \in \mathbb{Z}$. A homogeneous map $X \rightarrow Y$ of degree d is a map of degree zero $\tilde{X}\langle t+d \rangle \rightarrow \tilde{Y}\langle t \rangle$, so decomposing maps into homogeneous components gives (cf. [12, Lemma 2.1])

$$\mathbf{Hom}_R(X, Y) = \bigoplus_{s \in \mathbb{Z}} \mathbf{Hom}_{\mathbf{gr}}(\tilde{X}\langle s \rangle, \tilde{Y}\langle t \rangle). \quad (5.5)$$

For distinct underlying indecomposables every map is radical. For an endomorphism, a homogeneous component of nonzero degree is nilpotent, while its component of degree zero is radical exactly when its image in the residue field k is zero. Hence the radical of $\mathbf{mod} R$ between the underlying modules of graded lifts is a graded subspace, that is, it contains the homogeneous components of its elements. A map of degree zero between graded indecomposables is an isomorphism in $\mathbf{grmod} R$ exactly when its underlying map is an isomorphism, so the radical of $\mathbf{grmod} R$ consists of the maps of degree zero

in the radical of $\text{mod } R$. Therefore, under (5.5), the space $\text{rad}_R(X, Y)$ is the direct sum of the spaces $\text{rad}_{\text{grmod } R}(\tilde{X}\langle s \rangle, \tilde{Y}\langle t \rangle)$. A radical composite factors through a finite direct sum of indecomposable R -modules, each of which has a graded lift by Proposition 5.4. Composites of homogeneous maps are homogeneous, with degrees adding, so decomposing both factors into homogeneous parts shows that the same holds for the radical squares rad_R^2 and $\text{rad}_{\text{grmod } R}^2$. Consequently (cf. [13, Theorem 3.5])

$$\bigoplus_{s \in \mathbb{Z}} \text{Irr}_{\text{grmod } R}(\tilde{X}\langle s \rangle, \tilde{Y}\langle t \rangle) \cong \text{Irr}_R(X, Y). \quad (5.6)$$

We now restrict to the interval $[0, m]$ with $m \geq 3b$. Fix Y and a shift t with $b \leq t \leq m - 2b$. By (5.5) and the bound on degrees, a nonzero map $\tilde{Z}\langle u \rangle \rightarrow \tilde{Y}\langle t \rangle$ has $u \in [t - b, t + b]$, so its source is supported in $[t - b, t + 2b] \subseteq [0, m]$. In a radical factorization of a map into $\tilde{Y}\langle t \rangle$, we may omit the composites with a zero factor; each remaining intermediate object has a nonzero map to $\tilde{Y}\langle t \rangle$, so it is also supported in $[0, m]$. Since $\text{mod } R_m$ is a full subcategory of $\text{grmod } R$, its radical and radical square at the target $\tilde{Y}\langle t \rangle$ therefore agree with those of $\text{grmod } R$. By (5.6), the indegree of $\tilde{Y}\langle t \rangle$ in $\text{mod } R_m$ is $\text{indeg}_R(Y)$. There are $m - 3b + 1$ such shifts for each Y , and $\sum_Y \text{indeg}_R(Y) = a(R)$, so these targets contribute $(m - 3b + 1)a(R)$ arrows.

For each Y , at most $3b$ of the shifts t with $\tilde{Y}\langle t \rangle$ supported in $[0, m]$ lie outside $[b, m - 2b]$. At such a target, every irreducible space is a subquotient of a Hom space, and by (5.5) the total incoming Hom dimension is at most $\sum_X \dim_k \text{Hom}_R(X, Y)$, independently of t and m . These targets therefore contribute $O(1)$ arrows, which proves (5.3). Finally, Lemma 2.16 gives $\sigma(R_m) = 2(|\text{ind } R_m| - |R_m|) - a(R_m)$. Substituting (5.2), $|R_m| = (m + 1)|R|$ from Proposition 5.4, and (5.3) proves (5.4). $\square$

5.2. Positive gradings of Hom spaces. Interval algebras need not be representation-directed. The following grading of the Hom spaces between indecomposable modules ensures that they are.

Definition 5.6. A *positive Hom grading* on $\text{ind } R$ is a decomposition

$$\text{Hom}_R(X, Y) = \bigoplus_{d \geq 0} \text{Hom}^{(d)}(X, Y), \quad \text{Hom}^{(0)}(X, Y) = \begin{cases} k1_X & X = Y, \\ 0 & X \neq Y, \end{cases}$$

such that composition adds degrees.

Proposition 5.7. Let R be basic and representation-finite with a positive Hom grading.

(1) The grading induces a nonnegative grading on $R = \text{End}_R(R)$ and graded lifts

$$M_X = \text{Hom}_R(R, X), \quad (M_X)_d = \bigoplus_i \text{Hom}^{(d)}(e_i R, X).$$

(2) For these lifts,

$$\text{Hom}_{\text{gr}}(M_X\langle s \rangle, M_Y\langle t \rangle) = \text{Hom}^{(s-t)}(X, Y). \quad (5.7)$$

(3) Every graded indecomposable is a shift of one of the M_X , and $\text{grmod } R$ and every interval algebra R_m are directed.

Proof. (1) The grading extends to finite direct sums, hence to $\text{End}_R(\bigoplus_i e_i R) = R$, where $a \in R$ acts on R_R by left multiplication. Evaluation at 1 identifies $\text{Hom}_R(R, X)$ with X , with right action $(f \cdot a)(u) = f(au)$ for $u \in R$.

(2) A module map $M_X\langle s \rangle \rightarrow M_Y\langle t \rangle$ is postcomposition by a map $X \rightarrow Y$. Decomposing it into homogeneous components proves (5.7).

(3) Proposition 5.4 gives the classification up to shift. Each graded indecomposable has endomorphism ring k , and a nonzero map between distinct graded indecomposables strictly decreases the shift. There can be no cycle. Fullness of the interval subcategory proves the last assertion. $\square$

5.3. Standard forms and the lower bound. Proposition 5.7 requires a positive Hom grading. For a basic representation-finite algebra A , we obtain one on a standard form of A : an algebra with the same Auslander–Reiten quiver, whose category of indecomposable modules is the mesh category of that quiver (Definition 4.3).

Theorem 5.8 ([5, Sections 2.4, 2.8, 2.9, and 5.1]; [7, Section 3.1]). *Let A be a basic representation-finite algebra. There is a basic representation-finite algebra S with the following properties.*

- (1) *The Auslander–Reiten quivers of A and S are isomorphic, preserving arrow multiplicities, translation, and projective and injective vertices.*
- (2) *For a suitable mesh presentation of this quiver, the category of indecomposable S -modules is k -linearly equivalent to its mesh category, identifying vertices with the corresponding modules.*

We call any such algebra S a *standard form* of A . We use path length in a mesh presentation to grade the Hom spaces.

Construction 5.9. For a basic representation-finite algebra A , choose a standard form S , and a mesh presentation and an equivalence as in Theorem 5.8. For $X, Y \in \text{ind } S$, let $\text{Hom}^{(d)}(X, Y)$ be the image under this equivalence of the subspace spanned by paths of length d . Put $\text{Hom}^{(d)}(X, Y) = 0$ for $d < 0$.

The following lemma records the properties of these subspaces and the numerical comparison with A ; cf. [5, Section 5.1].

Lemma 5.10. (1) *The subspaces in Construction 5.9 give a grading*

$$\text{Hom}_S(X, Y) = \bigoplus_{d \geq 0} \text{Hom}^{(d)}(X, Y), \quad \text{Hom}^{(0)}(X, Y) = \begin{cases} k1_X & X = Y, \\ 0 & X \neq Y, \end{cases} \quad (5.8)$$

with composition adding degrees. The part of positive degree is the radical, and the components of degree at least two give its square. In particular, $\text{Hom}^{(1)}(X, Y) \cong \text{Irr}_S(X, Y)$.

- (2) *One has $|S| = |A|$ and*

$$\sigma(S) = \sigma(A). \quad (5.9)$$

Proof. (1) The mesh relations are homogeneous of degree two, so the path spaces give the grading (5.8). The part of positive degree is an ideal with quotient $\prod_X k1_X$, which is semisimple. It is nilpotent: there are finitely many finite-dimensional Hom spaces, so their degrees are bounded, and a composite of more maps of positive degree than this bound vanishes. It is therefore the radical. Every path of length at least two is a product of arrows, so those components give the radical square and the components of degree one give the irreducible spaces.

(2) Theorem 5.8(1) preserves the number of indecomposables, arrows, and indecomposable projectives; in particular $|S| = |A|$. Lemma 2.16 then gives (5.9). $\square$

By Lemma 5.10(1), the grading (5.8) is a positive Hom grading in the sense of Definition 5.6, so Proposition 5.7 applies to $R = S$.

Setting 5.11. For the remainder of this section and Section 6, choose a standard form S of a basic representation-finite algebra A , and fix the grading (5.8). Proposition 5.7 gives S a nonnegative grading and the graded lifts M_X of its part (1). By Lemma 5.10(1), $S^{(0)} = \bigoplus_i ke_i$ for the primitive idempotents e_i . There are finitely many finite-dimensional Hom spaces, so we may choose an integer $b \geq 1$ bounding all nonzero degrees in (5.8). In particular $S^{(d)} = 0$ for $d > b$.

By Propositions 5.4 and 5.7, all interval algebras S_m are representation-finite and representation-directed.

For the dual numbers we can describe the grading and all the interval algebras explicitly.

Example 5.12. Let $S = k[t]/(t^2)$. Its indecomposables are $P = S$ and $U = S/(t)$, with irreducible maps the quotient $\pi : P \rightarrow U$ and the socle inclusion $\iota : U \rightarrow P$. The cycle $P \rightarrow U \rightarrow P$ shows that S is not representation-directed, and Lemma 2.16 gives $\sigma(S) = 2(2 - 1) - 2 = 0$. Every morphism is a linear combination of identities, π , ι , and $\iota\pi$, subject to the mesh relation $\pi\iota = 0$. Thus the

indecomposable category is its mesh category, and S may be taken as its own standard form. Since $\iota\pi$ is multiplication by t , path length gives t degree two. Accordingly, M_P has one-dimensional components in degrees zero and two, while M_U is concentrated in degree one.

For $m \geq 2$, let Q_m be the quiver with vertices $0, \dots, m$ and arrows $r \rightarrow r+2$ for $0 \leq r \leq m-2$. Then S_m is the opposite of kQ_m modulo all paths of length two, so right S_m -modules are representations of Q_m , with arrows $r \rightarrow r+2$, in which all composites of two arrows vanish. Its indecomposables are the $m+1$ simples and the $m-1$ shifts of M_P supported in degrees r and $r+2$, for $0 \leq r \leq m-2$. Each nonsimple indecomposable has an irreducible map from its socle and one to its top. Hence

$$|S_m| = m+1, \quad |\text{ind } S_m| = 2m, \quad a(S_m) = 2(m-1), \quad \sigma(S_m) = 0.$$

Theorem 5.13. *Every representation-finite algebra A satisfies $\chi(\text{mod } A) \geq |A|$.*

Proof. By Lemma 2.18, we may assume that A is basic. Choose a standard form S . Since the interval algebras S_m are representation-finite and representation-directed, Corollary 4.26 gives $\sigma(S_m) \geq 0$ for every m . Divide (5.4) by $m+1$ and let $m \rightarrow \infty$. This gives $\sigma(S) \geq 0$, and (5.9) gives $\sigma(A) \geq 0$. $\square$

6. EQUALITY IN THE LOWER BOUND AND SPECIAL BISERIAL ALGEBRAS

Theorem 5.13 proves $\chi(\text{mod } A) \geq |A|$, but its asymptotic argument does not by itself determine when equality holds. We first bound, on each finite interval, a nonnegative invariant of interval algebras that does not increase under primitive deletion. We then show that the almost split sequences of a graded algebra occur in long interval algebras.

6.1. Monotone invariants of interval algebras. The following lemma applies to any invariant with properties (a)–(c) below; we use it for the surplus in Proposition 6.2. For an interval algebra R_m , call the algebras $R_m/R_m\eta R_m$, for idempotents η of the diagonal subalgebra $(R^{(0)})^{m+1}$, its *idempotent quotients*.

Lemma 6.1. *Let R be a nonnegatively graded finite-dimensional algebra with $R^{(d)} = 0$ for $d > b$. Let f be a real-valued invariant defined on the idempotent quotients of the interval algebras of R and on finite products of these algebras, and let λ be a real number, with the following properties.*

- (a) f is additive on products.
- (b) For every idempotent quotient C of an interval algebra and every primitive idempotent ε of the image in C of the diagonal subalgebra, $f(C/C\varepsilon C) \leq f(C)$.
- (c) $f(R_m) \geq 0$ for every m , and $f(R_m) = (m+1)\lambda + o(m)$ as $m \rightarrow \infty$.

Then, for every $n \geq 0$,

$$0 \leq f(R_n) \leq (n+b+1)\lambda.$$

In particular, $\lambda = 0$ forces $f(R_n) = 0$ for every n .

Proof. Fix n , put $\ell = n+b+1$, and for $q \geq 1$ put $M = q\ell - b - 1$. Consider the q intervals

$$[j\ell, j\ell + n], \quad 0 \leq j < q,$$

inside $[0, M]$, and delete the degrees in the gaps between them, that is, pass to the idempotent quotient by the sum of the diagonal idempotents of these degrees. Distinct intervals among them are separated by b degrees, so all blocks between them vanish. No map within one of these intervals factors through a deleted degree, since degrees are monotone under multiplication. The quotient is therefore $\prod_{j=0}^{q-1} R_n$. Deleting the corresponding primitive idempotents of the diagonal subalgebra one at a time passes through idempotent quotients of R_M . Monotonicity and additivity therefore give

$$qf(R_n) \leq f(R_M) = (q\ell - b)\lambda + o(q).$$

Divide by q and let $q \rightarrow \infty$. Nonnegativity gives the other bound. $\square$

We apply this lemma with $f = \sigma$ to the standard form S and the grading in Setting 5.11.

Proposition 6.2. *If $\sigma(S) = 0$, then $\sigma(S_n) = 0$ and S_n is biserial for every $n \geq 0$.*

Proof. All the interval algebras are basic, since their diagonal quotients are products of copies of $S^{(0)} = \bigoplus_i ke_i$. They are representation-finite and representation-directed by Propositions 5.4 and 5.7, and so are their idempotent quotients, by repeated application of Lemma 4.8; a primitive deletion of a basic algebra is basic. The hypotheses of Lemma 6.1 hold with $\lambda = \sigma(S)$: (a) by Lemma 2.18, (b) by Proposition 4.19, and (c) by Corollary 4.26 and Theorem 5.5(2). That lemma gives $0 \leq \sigma(S_n) \leq (n + b + 1)\sigma(S) = 0$, and Corollary 4.26 shows that S_n is biserial. $\square$

6.2. Almost split sequences of long interval algebras. We first define the number $\beta(C)$ that appears in the characterization of biserial algebras of Theorem 6.4, then prove that the almost split sequences of a graded algebra occur in finite intervals.

Definition 6.3. For a representation-finite algebra C , let $\beta(C)$ be the maximum number of nonprojective indecomposable summands in the middle term of an almost split sequence of right C -modules, counting multiplicities. An empty maximum is zero.

Theorem 6.4 ([2, Theorem 4.7]; [22, p. 172 and Corollary to Theorem 1, p. 173]). *A representation-finite algebra C is biserial if and only if $\beta(C) \leq 2$.*

The implication that C is biserial if $\beta(C) \leq 2$ is due to Auslander and Reiten [2, Theorem 4.7], and Skowroński and Waschbüsch prove the converse [22, p. 173]. In [22] the equivalence is stated with the bound on both left and right modules, that is, on $\beta(C)$ and $\beta(C^{\text{op}})$. By an observation of Gabriel recorded there [22, p. 172], $\beta(C) = \beta(C^{\text{op}})$ whenever one of them is at least three, so the bound on $\beta(C)$ suffices.

Proposition 6.5. *Let R be a nonnegatively graded representation-finite algebra. For all sufficiently large m ,*

$$\beta(R) \leq \beta(R_m).$$

Proof. By [13, Theorem 3.5], the category $\text{grmod } R$ has almost split sequences, and forgetting the grading sends them to almost split sequences of $\text{mod } R$. By Proposition 5.4, every indecomposable R -module has a graded lift, so every almost split sequence of $\text{mod } R$ arises in this way. A graded almost split sequence whose terms are supported in $[0, m]$ remains almost split in the full subcategory of graded modules supported in $[0, m]$, that is, in $\text{mod } R_m$; by [13, Lemma 3.6], each graded almost split sequence has such a shift for m large.

For each nonprojective middle summand, take the graded almost split sequence ending at that particular shift of the summand. It gives a nonsplit epimorphism onto the summand, so the summand is nonprojective in every interval containing the support of this sequence. There are only finitely many almost split sequences and summands to consider. Shift each almost split sequence of R , together with the sequences ending at its nonprojective middle summands, by the same integer so that the least degree occurring in this family is zero, and choose m_0 such that $[0, m_0]$ contains all their supports. For every $m \geq m_0$, every nonprojective middle summand counted over R is counted over R_m with the same multiplicity. Taking the maximum proves the inequality. $\square$

6.3. Algebras attaining the lower bound. We now show that an algebra with zero surplus is special biserial, by passing from the interval algebras of a standard form back to the algebra.

Proposition 6.6. *If A is basic and representation-finite and $\sigma(A) = 0$, then A is special biserial.*

Proof. Choose a standard form S of A . Then $\sigma(S) = 0$ by (5.9). By Proposition 6.2, $C = S_m$ is biserial for every m , so Theorem 6.4 gives $\beta(C) \leq 2$. Choose m sufficiently large for Proposition 6.5; it gives the same bound for S . The isomorphism $\Gamma_A \cong \Gamma_S$ in Theorem 5.8 preserves projective vertices and arrow multiplicities, so $\beta(A) = \beta(S) \leq 2$. Another application of Theorem 6.4 shows that A is biserial. It is therefore special biserial by Theorem 4.25. $\square$

Proposition 6.7 ([6, Theorem 3.5]). *If A is representation-finite and special biserial, then $\chi(\text{mod } A) = |A|$.*

Theorem 5.13 and Propositions 6.6 and 6.7 prove [6, Conjecture A] over an algebraically closed field.

Theorem 6.8. *Let A be a finite-dimensional representation-finite algebra over an algebraically closed field. Then the following hold.*

- (1) $\chi(\text{mod } A) \geq |A|$.
- (2) $\chi(\text{mod } A) = |A|$ if and only if A is special biserial.

Proof. Magnitude, $|A|$, and being special biserial are invariant under Morita equivalence. We may therefore assume that A is basic.

- (1) Theorem 5.13 proves the inequality.
- (2) The implication $\Rightarrow$ is Proposition 6.6, and $\Leftarrow$ is Proposition 6.7.

□

APPENDIX A. DIRECT PROOFS FOR IDEAL QUOTIENTS OF REPRESENTATION-DIRECTED ALGEBRAS

We work in Setting 4.7. This appendix proves directly the three results listed at the start of Section 4. Subsection A.1 proves Proposition 4.12. Subsection A.2 determines the arrows at τ -projectives and τ -injectives and constructs a function on the indecomposables of $\mathcal{D}$ as in Lemma 4.2, which Subsection A.4 uses. Subsection A.3 proves Proposition 4.15, and Subsection A.4 proves Corollary 4.17. The proofs use the evaluation functor $F(X) = Xe$, the τ -sequences of Proposition 3.7, and Auslander–Reiten duality, and do not use [21].

A.1. A bound on composition multiplicities. By (4.3), $d_Z = \dim_k \text{Hom}_A(Z, I)$. We therefore first bound $\dim_k \text{Hom}_A(Z, I)$ when $\text{Ext}_A^1(U, Z) = 0$ for every submodule $U \subseteq I$. The idea of counting isomorphism classes of kernels is taken from the argument in [21, Section 6]. Under this vanishing hypothesis it gives the following lemma.

Lemma A.1. *Let A be representation-finite and I_0 an injective A -module. Suppose $Z \in \text{mod } A$ satisfies $\text{End}_A(Z) = \text{End}_A(I_0) = k$ and $\text{Ext}_A^1(U, Z) = 0$ for every submodule $U \subseteq I_0$. Then $\dim_k \text{Hom}_A(Z, I_0) \leq 1$.*

Proof. Let $f, g : Z \rightarrow I_0$ be nonzero maps with isomorphic kernels. If the kernels are nonzero, apply $\text{Hom}_A(-, Z)$ to $0 \rightarrow \ker f \rightarrow Z \rightarrow \text{Im } f \rightarrow 0$. Since $\text{Ext}_A^1(\text{Im } f, Z) = 0$, an isomorphism $\ker f \rightarrow \ker g$, followed by inclusion in Z , extends to an endomorphism of Z . The extension is a nonzero scalar, so the two kernels are equal as submodules. They are also equal if both are zero. Then $\text{Im } f \cong Z/\ker f = Z/\ker g \cong \text{Im } g$, and this isomorphism extends to a nonzero scalar endomorphism of I_0 by injectivity. Thus g is proportional to f .

Different lines in $\text{Hom}_A(Z, I_0)$ therefore have nonisomorphic kernels. There are only finitely many kernel isomorphism classes: their lengths are bounded by the length of Z , and there are finitely many indecomposables. Since k is algebraically closed, it is infinite. A space of dimension at least two over k has infinitely many lines, a contradiction. □

At the τ -injectives of $\mathcal{D}$, Auslander–Reiten duality gives the vanishing of $\text{Ext}_A^1(U, Z)$ required in Lemma A.1; the τ -projectives follow by duality.

Direct proof of Proposition 4.12. Let Z be an indecomposable A -module. If Z is injective, or if $\tau_A^- Z$ belongs to $\text{mod } B$, then $\text{Ext}_A^1(U, Z) = 0$ for every $U \subseteq I$. In the second case, apply Auslander–Reiten duality (2.2); ordinary Hom already vanishes:

$$\text{Hom}_A(\tau_A^- Z, U) \hookrightarrow \text{Hom}_A(\tau_A^- Z, I) \cong D_k((\tau_A^- Z)e) = 0.$$

Since I is the indecomposable injective envelope of E , Lemma 4.9(1) gives $\text{End}_A(Z) = \text{End}_A(I) = k$. Lemma A.1 therefore applies and gives $d_Z = \dim_k \text{Hom}_A(Z, I) \leq 1$. The same bound holds when Z is projective or $(\tau_A Z)e = 0$. To see this, apply the argument to A^{op} , the same idempotent e , and the right A^{op} -module $D_k Z$. The duality D_k satisfies $(D_k Z)e \cong D_k(Ze)$, so it preserves dimension at e and sends $\text{mod } B$ to $\text{mod } B^{\text{op}}$. It preserves directedness, exchanges projectives with injectives, and satisfies $D_k \tau_A Z \cong \tau_{A^{\text{op}}}^- D_k Z$. Hence Z is projective or has $(\tau_A Z)e = 0$ exactly when $D_k Z$ is injective or has $(\tau_{A^{\text{op}}}^- D_k Z)e = 0$.

(1) By Lemma 4.11, every τ -injective of $\mathcal{D}$ is either injective over A or has its inverse translate in $\text{mod } B$, and dually for τ -projectives. The bound $d_Z \leq 1$ for such Z , and its dual, therefore give $d_X \leq 1$, and $d_X \geq 1$ for objects of $\mathcal{D}$ by Lemma 4.9(1).

(2) Suppose $(\tau_A Y)e = 0$. If $Ye \neq 0$, then Y is τ -projective in $\mathcal{D}$ by Lemma 4.11, so (1) gives $d_Y = 1$; if $Ye = 0$, then $d_Y = 0$. The second implication is dual. $\square$

A.2. Arrows at τ -projectives and the height function. The τ -sequences of $\mathcal{D}$ give the following identities. Ringel and Vossieck use the equations in part (2) [21, Section 1, p. 217].

Lemma A.2. *In Setting 4.7, the following hold.*

(1) If $\tau_{\mathcal{D}}^+ X \neq 0$, then for every $Y \in \text{ind } \mathcal{D}$,

$$\dim_k \text{Irr}_{\mathcal{D}}(Y, X) = \dim_k \text{Irr}_{\mathcal{D}}(\tau_{\mathcal{D}}^+ X, Y). \quad (\text{A.1})$$

(2) For $X \in \text{ind } \mathcal{D}$, evaluation at e satisfies

$$d_X - d_{\theta_{\mathcal{D}}^+ X} + d_{\tau_{\mathcal{D}}^+ X} = \mathbf{1}_{\{X=P\}}, \quad d_X - d_{\theta_{\mathcal{D}}^- X} + d_{\tau_{\mathcal{D}}^- X} = \mathbf{1}_{\{X=I\}}. \quad (\text{A.2})$$

Here $\mathbf{1}_{\{X=P\}}$ is one if $X \cong P$ and zero otherwise, and similarly for I .

Proof. (1) By Proposition 2.7(2), the right τ -sequence at X is also the left τ -sequence at $\tau_{\mathcal{D}}^+ X$, so $\theta_{\mathcal{D}}^-(\tau_{\mathcal{D}}^+ X) \cong \theta_{\mathcal{D}}^+ X$. Lemma 2.9, applied to both sequences, gives (A.1).

(2) The category $\mathcal{D}$ is a strict τ -category by Proposition 3.7. Evaluate the first exact sequence of Lemma 2.13 at P . Since $\mathcal{D}(P, Z) \cong Ze$ by (4.3), this gives $d_X - d_{\theta_{\mathcal{D}}^+ X} + d_{\tau_{\mathcal{D}}^+ X} = \dim_k S_X(P)$. By Lemma 4.9(1), $S_X(P)$ is k if $X \cong P$ and zero otherwise. The second identity in (A.2) follows in the same way, by evaluating the second sequence of Lemma 2.13 at I and using $\mathcal{D}(Z, I) \cong D_k(Ze)$. $\square$

With Proposition 4.12(1), proved in Subsection A.1, these identities determine the arrows at τ -projectives and τ -injectives. Ringel and Vossieck also observe this [21, Section 1, p. 218].

Corollary A.3. *In $\Gamma_{\mathcal{D}}$, each τ -projective other than P has exactly one incoming arrow, with multiplicity one; P has none. Dually, each τ -injective other than I has exactly one outgoing arrow, and I has none.*

Proof. For a τ -projective $X \neq P$, equation (A.2) and Proposition 4.12(1) give $d_{\theta_{\mathcal{D}}^+ X} = d_X = 1$. Every indecomposable summand of $\theta_{\mathcal{D}}^+ X$ has positive integral dimension at e , so there is exactly one, occurring once. Lemma 2.9 then shows that X has exactly one incoming arrow, of multiplicity one. For P , Lemma 4.9(1),(2) gives $d_P = \dim_k \text{End}_A(P) = 1$, so equation (A.2) gives $d_{\theta_{\mathcal{D}}^+ P} = 0$. Thus there are no incoming arrows. The second formula in (A.2) gives the outgoing counts by the same argument. $\square$

The following proposition constructs on $\Gamma_{\mathcal{D}}$ a function as in Lemma 4.2(2), and describes the nonzero maps of $\mathcal{D}$ by powers of the radical; Subsection A.4 uses both. Its proof uses Corollary A.3 and the arrow pairing (A.1).

Proposition A.4. *There is a unique function $h : \text{ind } \mathcal{D} \rightarrow \mathbb{Z}_{\geq 0}$, called the height, with $h(P) = 0$ such that every arrow $X \rightarrow Y$ satisfies $h(Y) = h(X) + 1$, and $h(X) = h(\tau_{\mathcal{D}}^+ X) + 2$ whenever $\tau_{\mathcal{D}}^+ X \neq 0$. For distinct X, Y with $\mathcal{D}(X, Y) \neq 0$, put $\ell = h(Y) - h(X)$. Then $\ell > 0$ and*

$$\mathcal{D}(X, Y) = \text{rad}_{\mathcal{D}}^{\ell}(X, Y), \quad \text{rad}_{\mathcal{D}}^{\ell+1}(X, Y) = 0. \quad (\text{A.3})$$

Put $L = h(I)$. All heights lie in $[0, L]$, with P, I the unique objects of heights 0, L , respectively.

Proof. Order the vertices so that the source of every arrow precedes its target; this is possible by directedness. We define the heights in that order. At P , set $h(P) = 0$. For a τ -projective $Q \neq P$, its unique incoming arrow $Y \rightarrow Q$ gives $h(Q) = h(Y) + 1$. If X is not τ -projective, then for each incoming arrow $Y \rightarrow X$, equation (A.1) gives $\text{Irr}_{\mathcal{D}}(\tau_{\mathcal{D}}^+ X, Y) \neq 0$, so there is an arrow $\tau_{\mathcal{D}}^+ X \rightarrow Y$. Both $\tau_{\mathcal{D}}^+ X$ and Y precede X , so the arrow $\tau_{\mathcal{D}}^+ X \rightarrow Y$ gives $h(Y) = h(\tau_{\mathcal{D}}^+ X) + 1$ by induction. Set $h(X) = h(\tau_{\mathcal{D}}^+ X) + 2$. Induction proves the asserted arrow and translation identities.

Choose maps representing bases of the irreducible spaces. By Lemma 2.10, every nonisomorphism is a linear combination of composites of these maps. Every path from X to Y has length $h(Y) - h(X)$,

so every map $X \rightarrow Y$ lies in $\text{rad}_{\mathcal{D}}^{h(Y)-h(X)}$ and no nonzero map lies in the next power; this proves (A.3). The nonzero maps $P \rightarrow X$ and $X \rightarrow I$ in (4.3) place all heights in the stated interval and give uniqueness at its endpoints. There is a path from P to every indecomposable. Since $h(P) = 0$ and height increases by one along each arrow, the height of each indecomposable is the length of any path from P to it. This proves uniqueness of h . $\square$

A.3. The equivalence with poset representations. Construction 4.14 describes the subspaces of Xe using maps from the τ -projectives. We prove fullness by induction and realize an arbitrary poset representation by taking a quotient of an explicit A -module.

Direct proof of Proposition 4.15. We divide the proof into three steps.

Step 1: The functor and its values on τ -projectives. Since F is faithful and $F(Q_t), F(Q_s)$ are one-dimensional, $\dim_k \mathcal{D}(Q_t, Q_s) \leq 1$. Composites of nonzero maps between these objects evaluate to composites of isomorphisms between lines, so they are nonzero. Thus the relation is transitive; identity maps give reflexivity, and directedness gives antisymmetry. Hence T is a poset. For $s \leq t$, normalize $w : Q_t \rightarrow Q_s$ so that $wu_t = u_s$. Then $fu_s = (fw)u_t$, proving $F_s(X) \subseteq F_t(X)$. Postcomposition by $a : X \rightarrow Y$ sends fu_t to $(af)u_t$, so G is a functor and is faithful.

Directedness gives $\mathcal{D}(Q_t, P) = 0$. Also $F_s(Q_t) = k$ exactly when $t \leq s$, giving the stated values on P, Q_t . Finally, $F(I) = Ie \cong D_k(eAe) = k$ and $\mathcal{D}(Q_t, I) \cong D_k(Q_te) \neq 0$, so $G(I) = K_T$. A map from $K_{\{s:t \leq s\}}$ to a T -space is determined by an arbitrary vector in its t -subspace. The map $\mathcal{D}(Q_t, X) \rightarrow F_t(X)$ is injective because $F(u_t)$ is an isomorphism and F is faithful, and it is surjective by definition. Maps from $K_\emptyset$ to $G(X)$ are arbitrary vectors of $F(X) = \mathcal{D}(P, X)$. Thus every map with source $G(P)$ or $G(Q_t)$ is induced by a map in $\mathcal{D}$, proving fullness for these sources.

Step 2: Fullness. Induct over sources in a topological order. For a non- τ -projective X , use its right τ -sequence $\tau_{\mathcal{D}}^+ X \xrightarrow{f} \theta_{\mathcal{D}}^+ X \xrightarrow{g} X$. Every summand of $\theta_{\mathcal{D}}^+ X$ precedes X . Moreover, $F(g)$ is surjective: every map $P \rightarrow X$ is a non-retraction and factors through g . Given $a : G(X) \rightarrow G(Y)$, induction lifts $aG(g)$ to $b : \theta_{\mathcal{D}}^+ X \rightarrow Y$. Faithfulness gives $bf = 0$. Since $\tau_{\mathcal{D}}^+ X \neq 0$, this is also a left τ -sequence, so g is a weak cokernel of f . Thus $b = vg$ for some $v : X \rightarrow Y$. Surjectivity of $F(g)$ gives $G(v) = a$. This proves fullness on indecomposables, and hence on direct sums.

Step 3: Essential surjectivity. For each τ -projective Q of $\mathcal{D}$ and each A -module V_0 with $V_0 = V_0 eA$, equivalently a quotient of a finite sum of copies of P ,

$$\text{Ext}_A^1(Q, V_0) = 0. \quad (\text{A.4})$$

Indeed this is immediate if Q is projective over A . Otherwise $\tau_A Q$ is annihilated by e , by Lemma 4.11. Since $V_0 = V_0 eA$, every map from V_0 to $\tau_A Q$ is zero: the generators in $V_0 e$ must map to $(\tau_A Q)e = 0$. Auslander–Reiten duality (2.2) gives $D_k \text{Ext}_A^1(Q, V_0) = \text{Hom}_A(V_0, \tau_A Q) = 0$.

Take a T -space $(W; W_t)$, and form the A -module

$$V = (P \otimes_k W) \oplus \bigoplus_{t \in T} (Q_t \otimes_k W_t).$$

Here A acts on the module factor. Our chosen bases identify $Ve = W \oplus \bigoplus_t W_t$. Let $\rho : Ve \rightarrow W$ be the sum of the identity on W and the inclusions $W_t \hookrightarrow W$, and put $N = \ker \rho$. Define

$$\alpha : P \otimes_k N \longrightarrow V, \quad \alpha(ea \otimes x) = xa, \quad V_0 = \text{Im } \alpha, \quad M = V/V_0.$$

The map α is well defined since $x \in Ve$. Since $eAe = k$, α_e is the inclusion $N \hookrightarrow Ve$, and exact evaluation gives $V_0 e = N$. Thus ρ identifies $Me = Ve/N$ with W , and the quotient map $Ve \rightarrow Me$ with ρ . Since $V_0 = V_0 eA$, we have $\text{Ext}_A^1(Q_s, V_0) = 0$ by (A.4). Applying $\text{Hom}_A(Q_s, -)$ to $0 \rightarrow V_0 \rightarrow V \rightarrow M \rightarrow 0$ shows that every map $Q_s \rightarrow M$ lifts to V . Representing morphisms of $\mathcal{D}$ by module maps therefore also gives a surjection $\mathcal{D}(Q_s, V) \rightarrow \mathcal{D}(Q_s, M)$. By Step 1, the P -summand contributes zero to $F_s(V)$, and the Q_t -summand contributes W_t exactly when $t \leq s$. Hence

$$F_s(V) = \bigoplus_{t \leq s} W_t \subseteq Ve, \quad F_s(M) = \rho(F_s(V)) = \sum_{t \leq s} W_t = W_s.$$

The equality $\sum_{t \leq s} W_t = W_s$ holds because every W_t in the sum is contained in W_s , and the term $t = s$ is present. Thus $G(M)$ is isomorphic to the given T -space $(W; W_t)$. Steps 1 and 2 prove

that G is fully faithful, and Step 3 proves essential surjectivity. Thus G is an equivalence. By its definition, T has one element for each τ -projective other than P , so $|T| = |\text{ind}_1^+ \mathcal{D}| - 1$. $\square$

A.4. Lower bound and equality for the magnitude of the quotient. By Lemma 4.2, the height function of Proposition A.4 determines the number of arrows of $\mathcal{D}$, and hence its magnitude.

Lemma A.5. *With $L = h(I)$, one has*

$$a(\mathcal{D}) = 2|\text{ind } \mathcal{D}| - L - 2, \quad (\text{A.5})$$

$$\chi(\mathcal{D}) - 1 = L - (|\text{ind}_1^+ \mathcal{D}| - 1). \quad (\text{A.6})$$

Proof. The category $\mathcal{D}$ is a strict τ -category by Proposition 3.7. Lemma 4.9(2) and Corollary A.3 give condition (1) of Lemma 4.2 with $X_0 = P$ and $X_1 = I$, and Proposition A.4 gives condition (2). Lemma 4.2 now gives both equations. $\square$

The proof of Corollary 4.17 below also uses two general facts. The first decomposes representations of posets of width at most two. It is well known; see [21, Section 8, Remark, p. 237] and [11, Section 5.2, Corollary].

Lemma A.6. *Let T be a finite poset whose antichains have at most two elements. Then every T -space is a direct sum of T -spaces with one-dimensional total space.*

Proof. We recall the usual argument. By Dilworth's theorem [10], T is a union of at most two chains. The subspaces of a T -space W along these chains, together with 0 and W , form two filtrations $0 = W'_0 \subseteq \cdots \subseteq W'_r = W$ and $0 = W''_0 \subseteq \cdots \subseteq W''_q = W$. Choose a complement C_{ij} of $(W'_{i-1} \cap W''_j) + (W'_i \cap W''_{j-1})$ in $W'_i \cap W''_j$. We show by induction on $i+j$ that $W'_i \cap W''_j = \bigoplus_{a \leq i, b \leq j} C_{ab}$. By induction, $W'_{i-1} \cap W''_j$, $W'_i \cap W''_{j-1}$, and their intersection $W'_{i-1} \cap W''_{j-1}$ are the direct sums of the C_{ab} over the corresponding index sets. Comparing dimensions shows that the sum of the first two is the direct sum of the C_{ab} with $a \leq i, b \leq j$, and $(a, b) \neq (i, j)$; adding C_{ij} completes the induction. Hence bases of the C_{ij} form a basis of W such that every indexed subspace is spanned by a subset of it. Each basis line, with its intersections with the indexed subspaces, is a T -space with one-dimensional total space, and W is their direct sum. $\square$

The second shows that a map formed from bases of the irreducible maps into an object is minimal right almost split. For module categories it follows from [3, VII, Lemma 1.2 and Proposition 1.3].

Lemma A.7. *Let $\mathcal{C}$ be a Hom-finite Krull-Schmidt k -linear category with finitely many indecomposable isomorphism classes. Let Y be indecomposable and $g : M \rightarrow Y$ a radical map such that, for each indecomposable Z , the components of g on the summands of M isomorphic to Z represent a basis of $\text{Irr}_{\mathcal{C}}(Z, Y)$. Then g is minimal right almost split.*

Proof. Evaluating at each indecomposable Z shows that P_g induces an isomorphism

$$P_M / \text{rad}_{\mathcal{C}}(-, M) \rightarrow \text{rad}_{\mathcal{C}}(-, Y) / \text{rad}_{\mathcal{C}}^2(-, Y).$$

Let $J \subseteq \text{rad}_{\mathcal{C}}(-, Y)$ be the image of P_g . Then $\text{rad}_{\mathcal{C}}(-, Y) = J + \text{rad}_{\mathcal{C}}^2(-, Y)$. Composing with radical maps gives $\text{rad}_{\mathcal{C}}^2(-, Y) \subseteq J + \text{rad}_{\mathcal{C}}^3(-, Y)$, and inductively $\text{rad}_{\mathcal{C}}(-, Y) = J + \text{rad}_{\mathcal{C}}^n(-, Y)$ for every n . The radical is nilpotent by Lemma 2.10, so $J = \text{rad}_{\mathcal{C}}(-, Y)$: every radical map into Y factors through g . Since g is radical, it is not a retraction, so g is right almost split. If $gu = g$ for some $u \in \text{End}_{\mathcal{C}}(M)$, then P_u induces the identity on $P_M / \text{rad}_{\mathcal{C}}(-, M)$. Evaluating at M gives $1_M - u \in \text{rad}_{\mathcal{C}}(M, M)$, so u is invertible. $\square$

Direct proof of Corollary 4.17. We divide the proof into four steps.

Step 1: $L \geq |T|$. By Lemma A.5 and the equality $|T| = |\text{ind}_1^+ \mathcal{D}| - 1$ of Proposition 4.15, $\chi(\mathcal{D}) - 1 = L - |T|$, so it remains to show $L \geq |T|$. Each K_U is indecomposable, since its endomorphism ring is k . We use the equivalence G to regard it as an object of $\mathcal{D}$ and write $h(K_U)$ for the height of that object. For upward-closed U, V , the space $\text{Hom}(K_U, K_V)$ is k when $U \subseteq V$, and zero otherwise. In the nonzero case take the map that is the identity on total spaces. Every upward-closed subset occurs in a chain from $\emptyset$ to T adding one element at a time. The inclusions in

such a chain are nonzero nonisomorphisms, so each raises height by at least one by Proposition A.4. There are $|U|$ steps before K_U and $|T| - |U|$ after it. Since $K_\emptyset = G(P)$ and $K_T = G(I)$, we obtain

$$h(K_U) \geq |U|, \quad L - h(K_U) \geq |T| - |U|.$$

Thus $L \geq |T|$, and hence $\chi(\mathcal{D}) \geq 1$. If $L = |T|$, then

$$h(K_U) = |U| \quad \text{for every upward-closed } U. \quad (\text{A.7})$$

In particular inclusions adding one element give irreducible maps, by (A.3).

Step 2: (1) $\Rightarrow$ (3). Assume (1), that is, $L = |T|$, so that (A.7) holds. We show that T has no three-element antichain. We use the following observation.

Claim. If indecomposables X, Z_1, Z_2, Y , with $Z_1 \not\cong Z_2$, form a commutative square of irreducible maps as below, then $X \cong \tau_{\mathcal{D}}^+ Y$.

$$\begin{array}{ccc} X & \xrightarrow{a} & Z_1 \\ c \downarrow & & \downarrow b \\ Z_2 & \xrightarrow{d} & Y \end{array} \quad ba = dc.$$

To prove the claim, complete the classes of b, d to bases of the incoming irreducible spaces at Y ; their sources are nonisomorphic, so these classes lie in different spaces. By Lemma A.7, representatives of these bases form a minimal right almost split map. Since minimal right almost split maps ending at Y are unique up to isomorphism, and μ_Y^+ is one of them by Proposition 2.7(1), we may take b, d as components of $g = \mu_Y^+ : \theta_{\mathcal{D}}^+ Y \rightarrow Y$. The nonzero tuple $(a, -c, 0, \dots)$ is killed by g , so it factors through its weak kernel $\tau_{\mathcal{D}}^+ Y$. Hence $\tau_{\mathcal{D}}^+ Y \neq 0$ and $\mathcal{D}(X, \tau_{\mathcal{D}}^+ Y) \neq 0$. Both objects have height $h(Y) - 2$. By Proposition A.4, a nonzero map between distinct indecomposables strictly raises the height, so

$$X \cong \tau_{\mathcal{D}}^+ Y. \quad (\text{A.8})$$

Suppose t_1, t_2, t_3 are pairwise incomparable in T , and put $U = \{s : t_1 < s \text{ or } t_2 < s \text{ or } t_3 < s\}$. All sets $U \cup S$, $S \subseteq \{t_1, t_2, t_3\}$, are upward-closed. The chosen inclusion maps are identities on total spaces, so their squares commute. For $j = 2, 3$, the four objects

$$K_U, \quad K_{U \cup \{t_1\}}, \quad K_{U \cup \{t_j\}}, \quad K_{U \cup \{t_1, t_j\}}$$

form such a square, with distinct middle objects. The two squares have common source K_U and distinct targets. Equation (A.7) makes their arrows irreducible, and (A.8) gives the same translate K_U for their distinct targets. This contradicts the injectivity of $\tau_{\mathcal{D}}^+$ (Proposition 2.7(3)).

Step 3: (3) $\Rightarrow$ (2). If T has width at most two, Lemma A.6 decomposes every T -space into one-dimensional summands, and the equivalence of Proposition 4.15 gives $d_X = 1$ for every indecomposable of $\mathcal{D}$.

Step 4: (2) $\Rightarrow$ (1). If all $d_X = 1$, equation (A.2) gives two middle summands at each non- τ -projective and one at each τ -projective other than P . Hence $a(\mathcal{D}) = 2(|\text{ind } \mathcal{D}| - |\text{ind}_1^+ \mathcal{D}|) + |\text{ind}_1^+ \mathcal{D}| - 1$, and Proposition 2.14 gives $\chi(\mathcal{D}) = 1$. $\square$

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